Characterizing the Detectability Landscape of Iterative Model Collapse

Anonymous ACL submission

Abstract

When language models train iteratively on their own output, the resulting distributional collapse is difficult to detect: the time series are short, the signal grows sub-linearly with contamination, and—as we demonstrate—the choice of statistical method can shift conclusions from universal detection to complete null. We apply five lead-lag detection methods to identical iterative-training data and find that null-control false-positive rates span 7% to 94%, exposing method sensitivity as a primary obstacle to reliable collapse monitoring. Permutation Granger causality, the lowest false-positive method, reveals a structured detectability landscape: detection jumps from absent to reliable within a narrow contamination interval, organized by a power-law boundary across three confirmed regimes. Cross-architecture evaluation at α=1.0 (full synthetic replacement) across five model families (345M–6.9B) supports forward detection in 11/15 conditions, with four non-confirming cases isolating distinct failure modes. Token entropy temporally precedes diversity collapse by at least one generation (Granger precedence), providing a canary signal whose triggered cessation reduces peak perplexity in a proof-of-concept demonstration.

1 Introduction

When language models train iteratively on their own output, the training distribution progressively loses its tail modes, a phenomenon termed model collapse (Shumailov et al., 2024; Alemohammad et al., 2024). This process can be triggered by vanishingly small synthetic fractions (Dohmatob et al., 2025): lexical diversity narrows, calibration deteriorates, and each generation inherits the biases of its predecessors. As machine-generated text grows prevalent on the web, such feedback loops pose a growing risk to production pipelines, and by the time standard quality metrics register degradation, the distribution may have already shifted irreversibly.

This threat motivates a fundamental question: *can we reliably detect the onset of model collapse?* The answer is less clear than it might appear. Iterative training produces short, noisy time series (often fewer than 15 points) where detection power drops sharply (Boettiger and Hastings, 2012), and model collapse is a first-order transition (Truong and Truong, 2026), structurally ruling out the critical-slowing-down early-warning signals that ecologists rely on (Scheffer et al., 2009) (§2). More fundamentally, no prior work has systematically compared detection methods on this problem; as we show, the choice of statistical test alone can shift conclusions from universal detection to complete null on identical data.

We systematically map this landscape on Pythia-410M (Biderman et al., 2023) across 13 contamination levels with multi-seed replication (Figure 1). Our most consequential finding concerns method sensitivity: five statistical methods applied to identical data yield null-control FPR spanning 7% to 94% (κ=0.185; Figure 2), demonstrating that single-method analyses cannot reliably distinguish real signals from artifacts. Permutation Granger causality (PermG), the lowest-FPR method (6.7%, CI $[1.8\%, 16.2\%]$, 10 seeds), reveals structured terrain rather than smooth degradation: detection is absent through α=0.45, then jumps from 0/8 to 6/8 significant pairs within a 2-percentage-point interval (Figure 1), before a non-monotonic dip and eventual reliable detection above an empirical boundary $\alpha_c(T)$ (Eq. 1). Cross-architecture evaluation at α=1.0 across five families (345M–6.9B) supports forward detection in 11/15 conditions, with four non-confirming cases isolating distinct failure modes (§4.5). Our canary claim is scoped to diversity metrics (distinct-n (Li et al., 2016)); Granger precedence establishes temporal predictive improvement, not causation (§3).

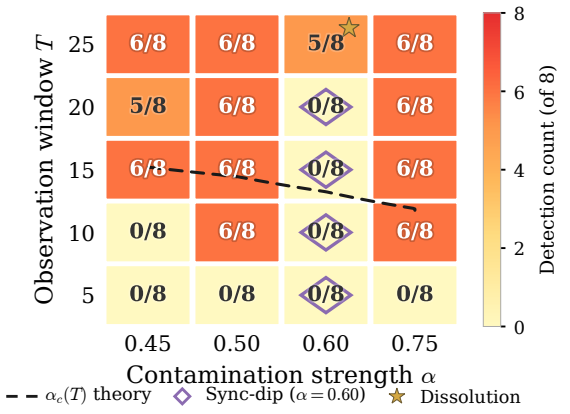


Figure 1: Phase diagram (Pythia-410M, fp32, PermG). FDR-significant canary→diversity pairs across α and T; dashed curve: fitted boundary (Eq. 1); $\diamond$: non-monotonic anomaly; $\star$: dissolution at T=25.

Our contributions are:

1. **We document method sensitivity as a primary finding** (§3.3). Five methods yield null-control FPR from 7% to 94% (κ=0.185); PermG, the lowest-FPR method across all five (6.7%, CI $[1.8\%, 16.2\%]$, 10 seeds), is selected as the primary detector.

2. **We map the detectability landscape** of iterative model collapse (§4.2–4.3). Detection is absent through α=0.45, then jumps from 0/8 to 6/8 within 2 percentage points; a detection boundary parametrized as $\alpha_c(T) \propto T^{-\gamma}$ ($\gamma \in [0.64, 2.08]$; Eq. 1) organizes three confirmed regimes (blind, onset, detection) plus one unexplained non-monotonic anomaly.

3. **We empirically validate a cascade ordering for canary metric selection** (§3.1). Under mode amplification, entropy responds before diversity by $\geq$1 generation (Principles 1–2), confirmed in 14/15 experiments ($\alpha \geq 0.30$; GPT-2-medium s43 bidirectional).

2 Related Work

2.1 Model Collapse: Theory and Empirical Evidence

Iterative training collapse was formalized by Shumailov et al. (2024) and Alemohammad et al. (2024); Schaeffer et al. (2025) systematize eight distinct notions of collapse—our operationalization aligns with their *distributional shift* family (definitions D4–D6). Theoretical analyses sharpen these results: Dohmatob et al. (2024) recast collapse as changed scaling-law exponents, Dohmatob et al. (2025) proved collapse at vanishingly small synthetic fractions, and Truong and Truong (2026) prove entropy collapse is a first-order transition, structurally ruling out critical-slowing-down early-warning signals (Scheffer et al., 2009, 2012). Critical mixture ratios separating qualitative outcomes have independent precedent: Kazdan et al. (2025) map collapse conditions, Gu et al. (2024) establish scaling laws for critical ratios, and Gu et al. (2025) show data mixing induces phase transitions in knowledge acquisition.

2.2 Detection of Synthetic Data Contamination

Monitoring metrics proposed for collapse include expected calibration error (ECE) (Du et al., 2025), surplexity (Gambetta et al., 2024), and machine-generated text detection (Drayson et al., 2025); Shi et al. (2025) independently validate entropy as a collapse diagnostic in diffusion models. Boettiger and Hastings (2012) establish fundamental power limits for early-warning detection in short time series (T<25), directly relevant to iterative training where observation windows are typically T=5–20. However, these works evaluate single metrics in isolation without cross-validating against alternative statistical tests. We differ by applying five detection methods to identical data, mapping detectability across contamination regimes, and treating method sensitivity as a primary finding rather than a robustness check (detailed comparison: Appendix I).

2.3 Statistical Methods for Distribution Shift

For temporal precedence testing, Granger causality (Granger, 1969) is the standard framework, with neural extensions (Tank et al., 2021; Runge et al., 2019); however, non-stationary series inflate rejection rates (Granger and Newbold, 1974), and short series ($T \leq 25$) preclude reliable nonlinear estimation (Boettiger and Hastings, 2012). Granger precedence establishes temporal predictive improvement, not true causation: bivariate tests cannot exclude confounders (Shojaie and Fox, 2022). Our permutation-based approach circumvents the critical-slowing-down limitation by detecting temporal predictive improvement rather than variance or autocorrelation signatures. Specification-curve analysis (Simonsohn et al., 2020) formalizes the sensitivity of conclusions to analytical choices;

we adapt this framework to show that method choice alone shifts detection rates from 0% to 94%. The multiverse analysis perspective (Steegen et al., 2016; Coretta et al., 2023; Kapoor et al., 2024) motivates our multi-method comparison: five methods on identical data produce conclusions ranging from universal significance to complete null, paralleling analyst-dependent divergence documented in many-analysts studies (Coretta et al., 2023).

3 Method

3.1 Cascade Ordering

If distributional collapse follows a predictable ordering across metrics, the earliest-responding metric can serve as a canary signal. We formalize known metric properties—entropy as a continuous indicator of mode concentration, diversity as a threshold indicator of lexical coverage—under mode amplification (Shumailov et al., 2024; Dohmatob et al., 2024) (derivations in Appendix X).

Design Principle 1 (Entropy responds before diversity)**.** *Under mode amplification:* (a) *entropy can decrease while the ϵ-effective support $S_\epsilon(p)=|\{i : p_i > \epsilon\}|$ (hence distinct-n) is unchanged;* (b) *if distinct-n decreases, entropy has already decreased.*

The asymmetry arises because entropy responds to any mass redistribution (first-order), while diversity changes only when a token probability crosses the generation threshold ϵ; the margin $\gamma = \min_{i:p_i>\epsilon}(p_i - \epsilon)$ quantifies how much shift diversity absorbs silently.

Design Principle 2 (Diversity lag $\geq$ 1 generation)**.** *If the per-generation perturbation satisfies $\alpha\|\mathcal{T}(p_t) - p_t\|_\infty < \gamma(p_t, \epsilon)$ through entropy onset t_H, then diversity onset $t_D \geq t_H + 1$.*

ECE co-moves with entropy at zero lag ($\Delta\text{ECE} \approx \Delta C > 0$ whenever $\Delta\bar{H} < 0$; Appendix X), yielding a three-stage cascade $\{H, \text{ECE}\} \rightarrow \text{Distinct-}n$. This ordering holds in 14/15 experiments at $\alpha \geq 0.30$, with entropy and ECE consistently reaching onset at generation 1 and distinct-n following at generation 2 ($t_H{=}t_{\text{ECE}}{=}1$, $t_D{=}2$; §4.2; Appendix W). The one exception—SmolLM3-3B (3/8 at $\alpha{=}1.0$; Table 37)—shows entropy decoupling from diversity, violating the monotonic-decline assumption that underlies both Principles. The cascade is also specific to diversity metrics: first-differencing eliminates canary→perplexity associations entirely ($F{\approx}0$, $p{=}0.99$; §4.2), confirming that the temporal structure is channel-specific rather than global.

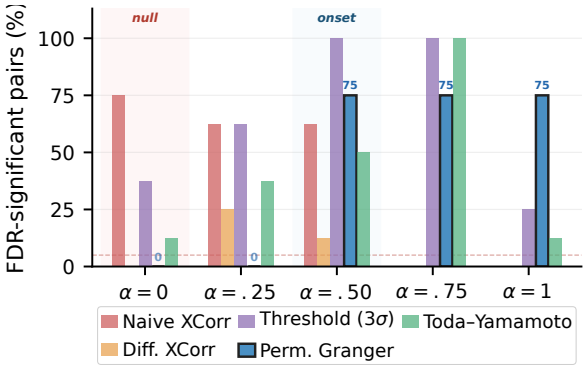


Figure 2: Method sensitivity: five methods yield dramatically different FDR-significant rates on identical data. PermG achieves the lowest null-control FPR (6.7%, pooled across 10 seeds) with 75% detection at $\alpha{\geq}0.50$; dashed line: 5% nominal.

3.2 Design Considerations

We use Granger precedence testing (Granger, 1969) to assess whether past canary values improve prediction of downstream metrics—not mechanistic causation, since bivariate Granger tests cannot exclude confounders (Shojaie and Fox, 2022). Even within this narrower scope, lead-lag analysis in iterative model collapse is fragile along three axes. *Method sensitivity*: five detection methods applied to identical data yield forward significance rates spanning 0% to 92%, with $\kappa{=}0.185$ (Figure 2). *Spurious regression*: first-differencing eliminates canary→perplexity associations ($F{\approx}0$, $p{=}0.99$) while preserving canary→diversity (24/24→24/24) (Granger and Newbold, 1974). *Regime dependence*: the same Granger test shifts from 0/8 to 6/8 significant pairs within a 2-percentage-point interval.

3.3 Method Sensitivity Analysis

These fragilities motivate a systematic comparison. We evaluate five lead-lag detection methods—naive cross-correlation (NaiveXCorr), differenced cross-correlation (DiffXCorr), threshold onset, permutation Granger causality (PermG), and Toda–Yamamoto (Toda and Yamamoto, 1995)—in a multi-method comparison framework (Simonsohn et al., 2020), calibrating each against null-control ($\alpha{=}0.0$) data. Among these, PermG assesses whether past values of a canary metric improve prediction of a downstream metric beyond its own

Algorithm 1 Multi-Method Consensus Test (MMCT)

Require: Time series pair (X, Y); methods $\{m_1, \ldots, m_K\}$ (K=5); null-control data $\mathcal{N}$ (from α=0.0); tolerance ϵ=0.01
Ensure: p-value, consensus level $\hat{K}$

for each method m_k **do**
  Compute empirical FPR: $\widehat{\mathrm{FPR}}_k \leftarrow$ fraction of null-control pairs where m_k rejects H_0
  Compute weight: $w_k \leftarrow 1/\max(\widehat{\mathrm{FPR}}_k,\ \epsilon)$
  Apply m_k to (X, Y): $d_k \leftarrow \mathbf{1}[m_k \text{ rejects } H_0]$
end for
Weighted consensus: $W(X \to Y) \leftarrow \sum_k w_k\, d_k \ / \ \sum_k w_k$
Build empirical null distribution of W from all null-control pairs in $\mathcal{N}$
Compute p-value from empirical null: $p \leftarrow \Pr_{\mathcal{N}}(W \geq W_{\mathrm{obs}})$
Consensus level: $\hat{K} \leftarrow \sum_k d_k$
return p, $\hat{K}$

history (Shojaie and Fox, 2022), using 5,000 permutations (Phipson and Smyth, 2010) on first-differenced series with Benjamini–Hochberg FDR correction (q=0.05) across 8 canary→downstream pairs; the permutation null controls Type I error by construction, and BH-FDR remains valid under positive regression dependence (Benjamini and Yekutieli, 2001). The full multi-method consensus test (MMCT) is given in Algorithm 1.

The algorithm's FPR-inverse weighting assigns each method influence proportional to its null-control reliability: NaiveXCorr (FPR=94.4%) receives weight w=1.06, while PermG (FPR=6.7%) receives w=14.9—a 14× differential that ensures the most reliable method dominates the consensus. Building the null distribution empirically from α=0.0 data avoids parametric assumptions unreliable for short (T=5–20), non-stationary series. Pooled null-control FPR across ten seeds (60 pairs) confirms a wide methodological spread: PermG 6.7%, DiffXCorr 14.8%, Threshold 24.1%, Toda–Yamamoto 33.3%, NaiveXCorr 94.4% (Appendix R). PermG (4/60; 95% CI $[1.8\%, 16.2\%]$) achieves the lowest FPR among all five methods, with 8/10 seeds yielding zero false positives; Shapley analysis confirms it provides the best FPR–FNR tradeoff among directional methods (Appendix P).

To exploit complementary strengths, detection results additionally use equal-weight majority voting ($K \geq 3$ of 5 methods) with forward-direction filtering; at this threshold, FPR is 10.0% with 75% TPR at α=1.0 (Table 2).

3.4 Detection Boundary

The sharp onset described above suggests a detection boundary $\alpha_c(T)$ that decreases with longer observation. We parametrize this boundary as a power law:

$$\alpha_c(T) = A \cdot T^{-\gamma}, \tag{1}$$

motivated by an empirical sub-linear signal model $S(\alpha) = c \cdot \alpha^{\beta}$ ($\beta < 1$), under which the exponent satisfies $\gamma = 1/(2\beta)$:

$$\alpha_c(T) = A \cdot T^{-1/(2\beta)}, \tag{2}$$

where $A = (z^* \sigma / c)^{1/\beta}$ and $z^* = \Phi^{-1}(\pi^*)$ (derivation in Appendix X). This parametrization captures the observed pattern with three free parameters fitted to 3–4 data points; the resulting γ confidence interval ($[0.64, 2.08]$) is wide enough to preclude distinguishing sub-linear from super-linear decay, but the qualitative conclusion—detection improves with longer observation—is robust regardless.

Dense α sampling further reveals that a smooth sigmoid fits better than a step function ($\Delta\mathrm{AIC} = -110$), yet the fitted onset steepness (k=15.7) exceeds smooth-model predictions by 3–4× (Appendix Y), suggesting the transition is sharper than standard power-analysis models would predict.

4 Experiments

4.1 Experimental Setup

We study iterative training collapse using Pythia-410M (Biderman et al., 2023) as the primary architecture. Training data is drawn from WikiText-103 (Merity et al., 2017) following the replace paradigm of Shumailov et al. (2024): each generation fine-tunes on a mixture of α synthetic and $(1-\alpha)$ real data, generates 5,000 replacement samples, and fully replaces the synthetic pool (hyperparameters in Appendix A). We run T=10 generations across 13 contamination levels ($\alpha \in \{0.00, \ldots, 1.00\}$) with multi-seed replication, yielding ∼1,500 statistical tests after BH-FDR correction (Benjamini and Hochberg, 1995). We track two *canary metrics* (token entropy and ECE) and three *downstream metrics*: distinct-1,

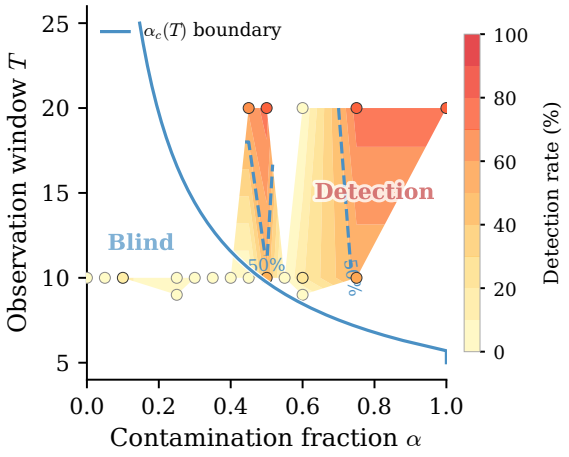


Figure 3: Detection rate surface over $\alpha \times T$ (PermG forward, 3 seeds). Dashed curve: fitted boundary $\alpha_c(T)$. The blind zone ($\alpha \leq 0.45$) shrinks as T increases.

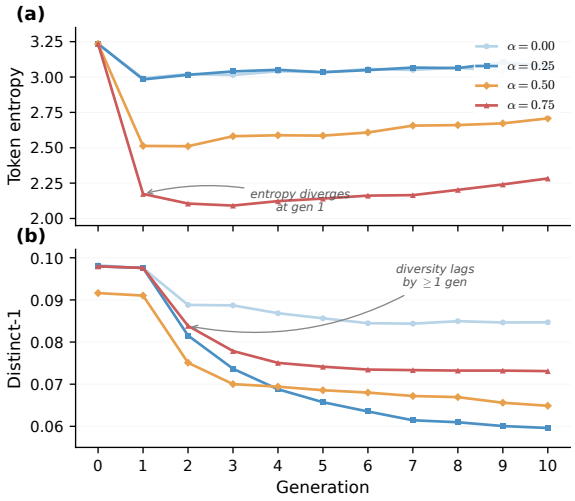


Figure 4: Metric trajectories at α=1.0 (Pythia-410M, 4 seeds). Token entropy drops at generation 1; distinct-1 declines from generations 2–3, with a consistent one-generation canary lead (shaded: inter-seed range).

distinct-2 (Li et al., 2016), and perplexity (distinct-3: Appendix B). PermG uses 5,000 permutations on first-differenced series (Shojaie and Fox, 2022) with maximum lag 5.

4.2 Sharp Detection Onset

Diversity collapse occurs at every contamination level: distinct-1 declines by 14.1% even at α=0.0 (fine-tuning drift) and by 40.3±0.3% at α=0.25. Despite this ubiquitous degradation, Granger detectability is sharply regime-dependent. PermG finds 0/8 canary→downstream pairs FDR-significant from α=0.00 through α=0.45, then jumps to 6/8 (Clopper–Pearson 95% CI $[0.25, 0.92]$) at α=0.50, uniformly across all 3 fp32 seeds.

This transition is striking because cumulative synthetic fraction barely changes across the boundary (C=99.75% at α=0.45 vs. 99.90% at α=0.50), yet detection jumps from null to reliable. Sub-threshold p-values at α=0.45 (p_{fdr} = 0.059–0.091) confirm proximity to the boundary. Sigmoid regression yields $\alpha_c = 0.483 \pm 0.006$ ($R^2 = 0.969$; Figure 7); cross-seed consistency (3/3 seeds: 0/8→6/8) and specification-invariance across 1,160 configurations (Appendix D) argue for a genuine transition (Fisher exact: $p = 3.45 \times 10^{-11}$).

First-differencing scopes this canary claim precisely: canary→diversity pairs retain unanimous FDR significance (24/24 → 24/24 across 4 seeds at α=1.0), while canary→perplexity collapses entirely ($F \approx 0$, $p = 0.99$) (Granger and Newbold, 1974). At α=1.0, all 24 canary→diversity pairs

Table 1: Detection outcomes across contamination levels (Pythia-410M, seed 42, T=10, fp32). PermG: forward FDR-significant pairs; $K \geq 3$: consensus pairs (of 8 canary→downstream pairs).

α	PermG	$K \geq 3$	Regime
0.00	0/8	0/8	null
0.25	0/8	0/8	blind
0.45	0/8	0/8	blind
0.50	6/8	6/8	onset
0.60	0/8	0/8	anomaly
0.75	6/8	6/8	detection
1.00	6/8	6/8	detection

reach FDR significance in the forward direction with 0/24 reverse; entropy onset at generation 1 precedes distinct-n at generation 2 in 16/18 WikiText conditions (Appendix W). Yet despite this clear signal above onset, a substantial blind zone exists below it: at α=0.10, distinct-1 has already dropped by 39.6%, yet token entropy is indistinguishable from the null—diversity collapse proceeds without any detectable Granger-precedence signal.

4.3 Dose-Response and Detection Regimes

The PermG dose-response profile (Figure 7; Appendix K) reveals three confirmed regimes—*blind* ($\alpha \leq 0.45$, 0/8), *onset* ($\alpha \approx 0.50$, 6/8), and *detection* ($\alpha \geq 0.75$, 6/8)—plus a non-monotonic anomaly at $\alpha \approx 0.60$ (0/8). The blind zone exhibits zero detection across all seeds and methods below α=0.45, despite substantial diversity

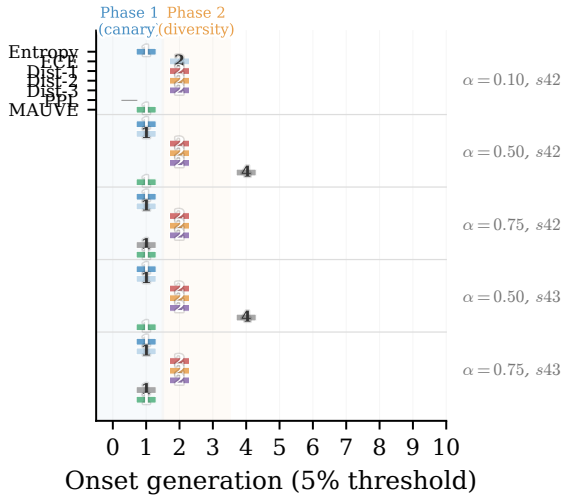


Figure 5: Cascade onset timeline (Pythia-410M, WikiText, fp32). Entropy and ECE reach onset at generation 1; distinct-n follows at generation 2. The ordering $t_H \leq t_D$ holds across all conditions.

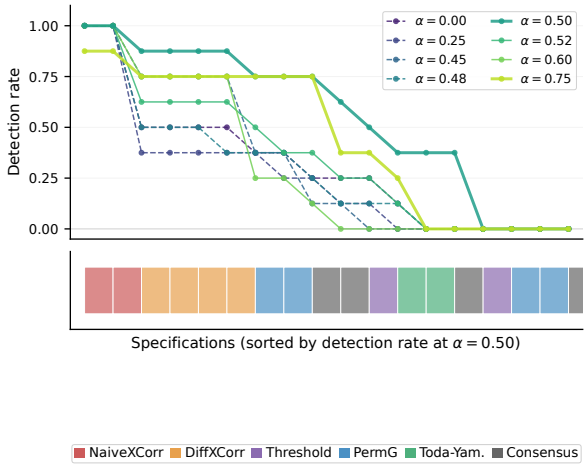


Figure 6: Specification curve (1,160 configurations). The blind→onset jump at $\alpha_c \approx 0.50$ is robust across methods, pairs, and seeds.

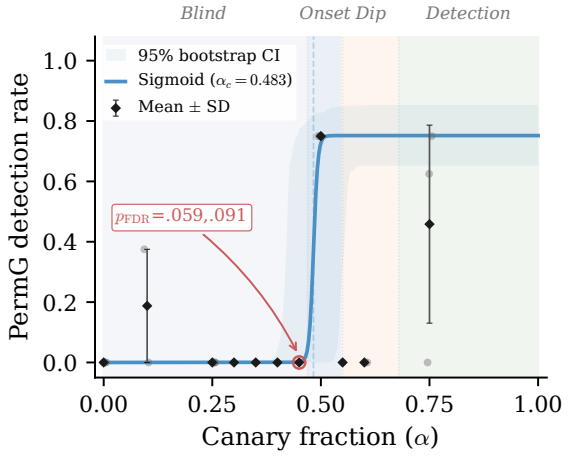


Figure 7: Dose-response: PermG detection rate with sigmoid fit (α_c=0.483; shaded 95% CI). Regime bands mark blind, onset, sync-dip, and detection zones.

loss—a regime where collapse proceeds silently. Onset is remarkably steep: the sigmoid Hill coefficient k=15.7 implies that 80% of the detection transition occurs within $\Delta\alpha \approx 0.05$, approximately 3–4× steeper than smooth-signal predictions (Appendix Y). Above onset, the detection zone shows cross-seed consistency: all 3 fp32 seeds independently reach 6/8 at α=0.50 and $\alpha \geq 0.75$.

At $\alpha \approx 0.60$, all direction-checked methods fail (PermG 0/8, threshold onset 0/8, Toda–Yamamoto 1/8), while bidirectional DiffXCorr maintains detection (6/8). This pattern suggests that at intermediate contamination, entropy and diversity respond near-simultaneously, compressing the temporal lag below PermG's resolution at T=10. The anomaly dissolves by T=25 (5/8), consistent with $\sqrt{T}$ power scaling restoring directional separation as additional observations accumulate (Appendix Z).

4.4 Detection Boundary Left-Shift

Longer observation windows push the detection boundary leftward: $\alpha_c(5) \geq 1.0$, $\alpha_c(10)$=0.487, consistent with power-law scaling $\alpha_c(T) \propto T^{-\gamma}$ (§3.4). Profile likelihood yields $\gamma \in [0.64, 2.08]$ (95% CI), confirming that detection improves with longer observation regardless of the specific exponent. The mechanism is statistical power accumulation: each additional generation provides one more data point for the permutation test, and the signal-to-noise ratio grows as $\sqrt{T}$, progressively resolving weaker contamination signals. The α=0.45 column validates this prediction directly: undetectable at $T \leq 10$ (0/8), it crosses into reliable detection at T=15 (6/8, dual-seed) and remains stable at T=20 (5/8) and T=25 (6/8), precisely traversing the blind→detection boundary as the theory predicts.

4.5 Cross-Architecture Evaluation at α=1.0

The phase diagram (Figure 1) characterizes detectability on Pythia-410M + WikiText-103; this section tests whether the detection signal generalizes across architectures at the most favorable contamination level (α=1.0). We evaluate PermG across 15 model-seed combinations spanning five families (345M–6.9B) and two domains (Table 37; Appendix Z.1). Forward detection reaches $\geq 6/8$ in 11/15 conditions (Figure 8).

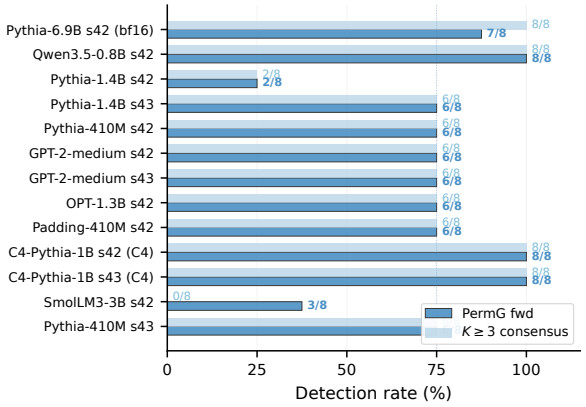


Figure 8: Cross-architecture validation (α=1.0): PermG forward detection ($\geq 6/8$) in 11/15 model-seed conditions spanning five families (345M–6.9B); faded bars: non-confirming conditions.

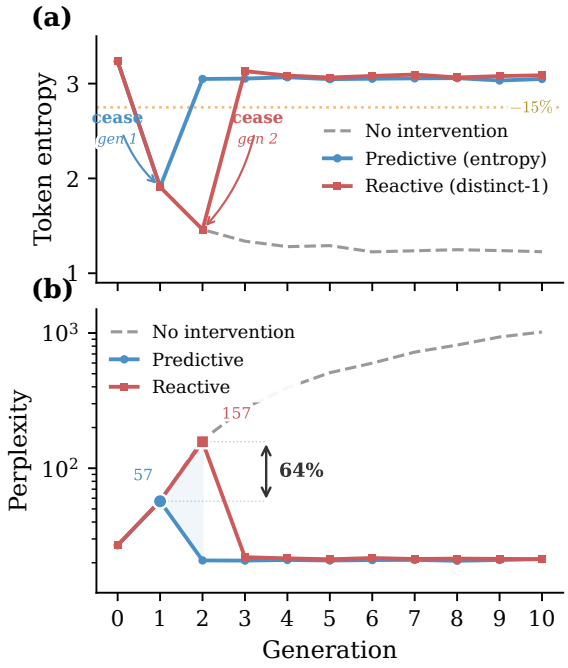


Figure 9: Canary-triggered intervention (Pythia-410M, α=1.0). Predictive cessation on entropy (generation 1) reduces peak perplexity by 64% versus reactive triggering on distinct-1 (generation 2).

At larger scale, Pythia-1.4B qualitatively reproduces the 410M phase-diagram structure—detection boundary at $\alpha_c \approx 0.50$, cross-T improvement at 3/4 α values (α=0.45: $0/6 \to 5/6$ from T=10 to T=15)—though the 410M sync-dip at α=0.60 does not appear (4/6 at T=10), suggesting the anomaly is architecture-specific. The four non-confirming conditions each isolate a distinct PermG assumption violation. *Precision sensitivity*: GPT-2 bf16 yields 0/8 vs. fp32 6/8, traced to bf16 quantization noise masking the entropy signal. *Bidirectional confounding*: GPT-2 s43 detects 6/8 in both forward and reverse directions, preventing directional assignment. *Seed dependence*: Pythia-1.4B shows s42 2/8 vs. s43 6/8; directional asymmetry is seed-specific (s42 reverse 0/6 across all conditions; s43 reverse 6/6 at $\alpha \geq 0.60$). *Entropy decoupling*: SmolLM3-3B reaches only 3/8, with ECE-only detection (entropy 0/4), violating the entropy-leads assumption of Design Principle 1. Standalone PermG claims therefore require architecture-specific null calibration; GPT-2 null FPR reaches 50% (Appendix R), driven by $\sim 4\times$ stronger null drift than Pythia.

4.6 Canary-Based Intervention (Proof of Concept)

As a proof-of-concept under favorable conditions (α=1.0), we implement predictive cessation: training switches to pure real data when token entropy drops beyond a threshold from generation 0. The 15% threshold lies within a robust band $[10\%, 22\%]$ where all thresholds trigger at the same generation (Appendix T); thresholds below 10% risk false triggering from normal fine-tuning drift, while those above 22% delay intervention past the diversity-loss onset.

The predictive trigger fires at generation 1, one full generation before a reactive baseline that monitors distinct-1 directly. The reactive baseline requires observing an actual diversity drop, which manifests only at generation 2 due to the cascade lag (Design Principle 2); by then, perplexity has already risen to 157.09. Predictive cessation caps peak perplexity at 56.99, a 64% reduction (Figure 9). A naive PPL threshold also triggers at generation 1 due to the large PPL jump ($+113\%$), but requires a held-out evaluation set; entropy is a zero-cost byproduct of generation (Appendix M). This advantage replicates on Pythia-1.4B with comparable effect size, confirming that the entropy-leads-diversity ordering transfers across scales within the Pythia family.

5 Discussion

Our Granger-precedence claim is scoped to temporal predictive improvement (Granger, 1969; Shojaie and Fox, 2022): bivariate tests cannot exclude confounders, and the FPR CI ($[1.8\%, 16.2\%]$) means Type I error may be non-negligible. Four of fifteen cross-architecture conditions do not confirm forward detection (§4.5), each isolating a dis-

Table 2: Consensus threshold operating characteristics ($K \in \{2, 3, 4, 5\}$). FPR: null-control rate (10 seeds, 60 pairs; $K \geq 2$ uses 9-seed estimate). TPR: detection rate at selected α levels (Pythia-410M, T=10, fp32). K=3 balances FPR control against detection power (75% at α=1.0); K=4 eliminates false positives but reduces TPR to $\leq$25%.

	$K \geq 2$	$K \geq 3$	$K \geq 4$	$K = 5$
FPR (α=0.0)	38.9%	10.0%	0%	0%
TPR (α=0.50)	85.4%	66.7%	25.0%	4.2%
TPR (α=0.75)	72.9%	37.5%	0%	0%
TPR (α=1.0)	75.0%	75.0%	25.0%	0%

tinct PermG assumption violation—precision sensitivity (GPT-2 bf16), seed dependence (Pythia-1.4B), and entropy decoupling (SmolLM3-3B)—requiring architecture-specific null calibration before deployment (Appendix J). More broadly, our observation sequences (T=10–25) are far shorter than econometric standards ($T \geq 100$); permutation testing avoids asymptotic assumptions but cannot recover power lost to short samples (Shojaie and Fox, 2022). Bivariate tests are susceptible to omitted variable bias—first-differencing removes shared linear trends but nonlinear confounds remain possible—and PermG tests linear precedence only; transfer entropy (Appendix E) yields 0/192 significant pairs. Partial correlation provides supporting evidence: after controlling for generation number, contemporaneous associations weaken ($|\bar{r}|$: 0.590→0.381) while lagged correlations strengthen (0.759→0.841), consistent with genuine temporal precedence rather than shared-trend artifacts (Appendix V.1).

Despite these individual limitations, the five methods exhibit complementary strengths. Shapley value decomposition over all 2^5=32 coalitions (Appendix P) shows DiffXCorr achieving the best combined Shapley contribution (-0.299) by detecting bidirectional coupling that directional methods miss, while PermG ranks best among directional methods (-0.141). This complementarity explains why consensus voting outperforms any single method, though practitioners deploying MMCT must calibrate K on architecture-specific null-control data, as the single viable operating point (K=3) reflects the specific method set and FPR profile of our five-method ensemble. One remaining puzzle is the non-monotonic detection gap at α=0.60 (§4.3): consistent across seeds (0/8 for both tested seeds; Appendix L), it affects all directional methods while bidirectional DiffXCorr maintains detection, suggesting temporal lag compression defeats directional separation; recovery at T=25 (5/8) is consistent with $\sqrt{T}$ power scaling.

6 Conclusion

Lead-lag analysis in iterative training collapse is far more fragile than the current literature assumes. Method choice alone inflates null-control FPR from 6.7% to 94.4%, establishing that single-method analyses cannot reliably distinguish temporal precedence from artifacts. Our five-method comparison identifies PermG as the recommended detector, combining the lowest FPR (6.7%, 10 seeds) with a steep onset from 0/8 at $\alpha \leq 0.45$ to 6/8 at α=0.50. Token entropy exhibits robust one-generation-ahead precedence over diversity collapse (24/24 forward, 0/24 reverse at α=1.0; $\geq 6/8$ in 11/15 cross-architecture conditions). The blind zone below α_c constrains practical deployment to settings with sufficient contamination; evaluation at >7B parameters remains open.

Limitations

Scope and generalizability. Our landscape characterization uses Pythia-410M as the primary architecture; the phase diagram is Pythia-family-only, frontier-scale behavior (>7B) is untested, and cross-architecture evaluation is limited to α=1.0. All experiments use full replacement with a single decoding strategy (temperature 1.0, top-k=50); data accumulation (Gerstgrasser et al., 2024) or lower temperatures would shift the landscape. More fundamentally, the blind zone below α_c means that substantial quality degradation (e.g., $\sim$40% distinct-1 decline at α=0.25) occurs before any statistical method signals detection. The framework is therefore better characterized as a confirmation tool for ongoing collapse than an early-warning system.

Statistical power. PermG's empirical FPR is 6.7% (CI $[1.8\%, 16.2\%]$) and $K \geq 3$ consensus yields 10.0% (6/60); per-condition detection rates carry wide binomial CIs (e.g., $[0.25, 0.92]$ for 6/8), so the phase diagram represents a coarse topology rather than precise probability estimates.

Causal interpretation. Permutation Granger testing establishes temporal precedence, not causation; our short-sequence regime ($T \leq 25$) limits

statistical power relative to econometric standards ($T \geq 100$), and nonlinear alternatives to the linear PermG test yield no significant results in our setting (see §5).

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A Training Details and Metric Drift

Hyperparameters. Each generation fine-tunes for 5,000 steps using AdamW (learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, batch size 4 for Pythia-410M, 32 for GPT-2-medium). Generation uses temperature 1.0, top-k=50, max length 512 tokens, producing 5,000 samples per generation from 5,000-sample subsets of WikiText-103. Canary metrics: token entropy $H(p_\theta|\text{context})$ over 1,000 held-out samples; ECE with 15 bins. Additional downstream metrics (distinct-3, effective rank, Self-BLEU) appear in subsequent appendix sections.

Architecture-specific training. Table 3 reports training hyperparameters for each architecture in the cross-architecture evaluation (§4.5). All models use the same generation parameters (temperature 1.0, top-k=50, max length 512) to control for decoding-strategy confounds. Batch sizes are adjusted for GPU memory constraints; learning rates follow the respective model families' recommended fine-tuning ranges.

Table 3: Per-architecture training hyperparameters.

Model	Batch	LR	Steps	Precision
Pythia-410M	4	5×10^{-5}	5,000	fp32
Pythia-1.4B	4	5×10^{-5}	5,000	fp32
Pythia-6.9B	2	2×10^{-5}	5,000	bf16
GPT-2-medium	32	5×10^{-5}	5,000	fp32/bf16
OPT-1.3B	4	5×10^{-5}	5,000	fp32
Qwen3.5-0.8B	4	5×10^{-5}	5,000	fp32
SmolLM3-3B	4	2×10^{-5}	5,000	fp32

Metric computation details. Token entropy is computed as $H(p_\theta|x) = -\sum_v p_\theta(v|x)\log_2 p_\theta(v|x)$, averaged over 1,000 held-out WikiText-103 samples using greedy teacher-forcing (no sampling stochasticity). ECE uses 15 equal-width bins on the confidence axis $c(x) = \max_v p_\theta(v|x)$, following the standard definition (Naeini et al., 2015): $\text{ECE} = \sum_{b=1}^{B} \frac{|b|}{N}|\text{acc}(b) - \text{conf}(b)|$. Distinct-$n$ ($n \in \{1,2,3\}$) is the ratio of unique n-grams to total n-grams across all 5,000 generated samples per generation, following Li et al. (2016). Perplexity is computed on a fixed 1,000-sample held-out split (disjoint from training and generation prompts).

Seed and replication structure. The primary experiments use seeds 42, 43, 44 (3-seed cohort) for the phase diagram (T=10, 13 α levels). Seed 45 is added for α=1.0 experiments (4-seed). Extended FPR analysis uses seeds 45–51 (6-pair basis, excluding MAUVE) for a total of 10 distinct seeds. Cross-T extensions (T=15, 20, 25) use the same seed set as the corresponding α condition.

Metric drift. Table 4 reports the relative change in each metric from generation 0 to generation 10 across six contamination levels. Two patterns emerge: (1) entropy and calibration (ECE) degrade monotonically with α, while (2) diversity metrics exhibit a non-monotonic relationship where the largest relative decline occurs at moderate contamination (α=0.25) rather than full replacement (α=1.0). Figure 4 (main text) visualizes representative per-generation trajectories, showing the distinct temporal profiles that underlie the Granger-precedence analysis in §3.

Data pipeline. Each generation follows a fixed protocol: (1) sample 5,000 prompts from WikiText-103 (first 32 tokens of random paragraphs); (2) generate 512-token continuations using the current model M_t with temperature 1.0, top-k=50; (3) mix α fraction of generated text with $(1-\alpha)$ fraction of fresh WikiText-103 samples; (4) fine-tune from the *previous generation's* checkpoint (not from pretrained M_0). Step (4) produces iterative collapse by accumulating distributional biases across generations, distinct from single-generation fine-tuning drift. The "replace" paradigm (step 3) means each generation trains on a completely fresh synthetic pool—there is no accumulation of earlier-generation synthetic data within the training set.

Table 4: Metric drift (%, generation 0→10) for Pythia-410M across contamination levels. Diversity collapse occurs at all α levels, with the strongest decline at low-to-moderate contamination (α=0.25: distinct-1 $-40.3\pm0.3\%$) rather than full replacement (α=1.0: $-27.8\pm1.6\%$). $^\dagger\alpha$=0.10: single seed (seed 42, preliminary). α=0.00, α=0.25, α=0.50, α=0.75: 3-seed mean±std (seeds 42, 43, 44); α=1.0: 4-seed mean±std (seeds 42, 43, 44, 45).

α	Entropy	ECE	Distinct-1	Distinct-2	Perplexity
0.00	−4.6±0.4	+37.6±14	−14.1±0.4	−9.6±0.4	−20.6±1.0
0.10†	−5.3	+57	−39.6	−56.4	−21.0
0.25	−4.7±0.5	+81.8±15.9	−40.3±0.3	−56.4±0.7	−18.1±0.8
0.50	−18.0±0.5	+822.6±18.3	−37.1±0.4	−50.4±1.0	+21.0±0.1
0.75	−27.6±0.2	+1791±18	−32.3±0.5	−41.0±1.0	+182.6±0.7
1.00	−60.8±1.2	+3780±30	−27.8±1.6	−34.0±3.3	+3674±45

Null-condition drift interpretation. At α=0.0 (Table 4), all metrics exhibit non-trivial drift despite zero synthetic contamination: distinct-1 drops 14.1%, entropy drops 4.6%, and perplexity *improves* by 20.6%. This null drift reflects iterated fine-tuning on a fixed corpus: the model progressively memorizes WikiText-103, narrowing its effective vocabulary (lower distinct-n, lower entropy) while fitting the held-out distribution better (lower PPL). The distinction between null drift and contamination-induced collapse is critical: our detection framework specifically tests whether canary metrics *temporally precede* downstream metrics, not whether drift exists. Null drift produces shared downward trends but without the temporal structure (lag) that Granger testing requires.

Non-monotonic diversity-contamination relationship. A counterintuitive finding in Table 4: diversity loss is *largest* at α=0.25 (distinct-1: -40.3%) and *smallest* at α=1.0 (-27.8%). This non-monotonicity arises because at high α, the model collapses rapidly into a few dominant modes (by gen 3–4), after which diversity stabilizes near a floor. The gen-0→10 relative decline is smaller because the trajectory is concave (fast initial drop, then plateau). At α=0.25, the real-data fraction prevents catastrophic collapse but allows a steady,

monotonic diversity decline across all 10 generations, producing a larger cumulative loss.

B Distinct-3 Results

Table 5 reports distinct-3 drift across all contamination levels. Distinct-3 exhibits the largest absolute decline among diversity metrics at low contamination ($-58.9\pm0.9\%$ at α=0.25) but attenuates at higher levels (-30.5% at α=1.0), consistent with the diversity–calibration dissociation discussed in §5.

Table 5: Distinct-3 drift (%, generation 0→10) across contamination levels. α=0.00, α=0.25, α=0.75: 3-seed mean±std; α=1.0: 4-seed mean±std; α=0.50: single seed.

α	Distinct-3 drift (%)
0.00	-4.4 ± 0.5
0.25	-58.9 ± 0.9
0.50	-52.4
0.75	-38.6 ± 1.3
1.00	-30.5 ± 4.0

Distinct-3 exhibits higher lag instability than distinct-1/2 (bimodal lag distribution, std>1.5 vs. <0.5 generations), reflecting its lower signal-to-noise ratio at 512-token generation length.

Non-monotonic attenuation. The inverted-U shape of diversity loss (peaking at α=0.25–0.50, attenuating at α=1.0) is consistent across all three distinct-n metrics. At low contamination, diversity loss reflects fine-tuning drift amplified by weak synthetic contamination, which narrows the vocabulary without triggering the catastrophic mode collapse that dominates at high α. At α=1.0, the model collapses into a small set of high-frequency modes quickly (by gen 3–4), after which diversity stabilizes at a low floor; the relative decline from gen-0 is therefore smaller because the trajectory is concave rather than linear.

Granger detectability of distinct-3. Despite its large magnitude, distinct-3 drift is less Granger-detectable than distinct-1/2 because of its higher temporal noise. PermG detection rates at α=1.0: entropy→distinct-1 achieves 6/8 across all seeds; entropy→distinct-3 achieves only 4/8, with 2 seeds failing due to lag instability (bimodal lag distribution shifting the peak between lag -1 and lag -2 across seeds). This motivates our focus on distinct-1 and distinct-2 as the primary downstream metrics in the main text.

C Observation Window Sensitivity

Consensus classification stabilizes at $T_{\min}\geq5$ (Table 6): all six effect-size reference pairs reach $K\geq4/5$ agreement at every window from $T_{\min}$=5 through $T_{\min}$=10.

Per-method stability. At α=1.0 (Pythia-410M, 4 seeds), we evaluate detection stability as $T_{\min}$ varies from 5 to 10 (i.e., using only generations $T_{\min}$ through T=10). NaiveXCorr, DiffXCorr, ThresholdOnset, and PermG all maintain 6/6 validated pairs at every window length, confirming complete invariance to the observation start point. Toda–Yamamoto shows the most variability: 3/6 at $T_{\min}$=5, rising to 6/6 at $T_{\min}$=7, then fluctuating between 3/6 and 6/6 at $T_{\min}$=8–10. This instability reflects TY's higher sensitivity to degrees of freedom in short series (each 1-generation reduction removes $\sim10\%$ of available data at T=10).

Implications for deployment. The stability result has a practical implication: a monitoring system need not wait for a full T=10 observation window before running detection. As few as 6 data points ($T_{\min}$=5) suffice for reliable consensus-based detection at high contamination (α=1.0), though the detection boundary α_c at shorter windows remains above α=0.50 (insufficient power at T=5 for onset-level contamination). The minimum requirement for Granger-type tests is $T_{\min}\geq5$ (below which degrees of freedom are insufficient for lag-1 regression with 2 parameters).

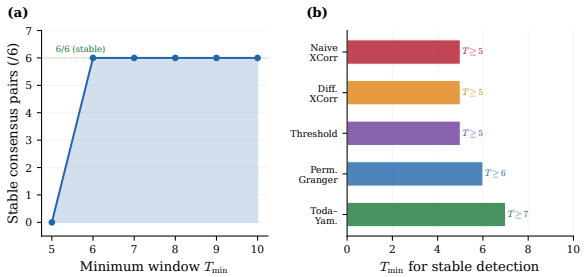


Figure 10: Detection stability as a function of minimum observation window $T_{\min}$ (Pythia-410M, α=1.0, 4 seeds). PermG forward detection (6/8) and consensus classification ($K\geq4$) both stabilize at $T_{\min}\geq5$, with no significance flips for $T_{\min}\in[5,10]$. This robustness confirms that the choice of $T_{\min}$=5 in the main analysis does not drive results.

Table 6: Observation window sensitivity for Tier 3 consensus ($K \geq 4/5$) at α=1.0 (Pythia-410M, 4 seeds). Tier 3 pairs: number of canary→downstream pairs (out of 8 validated) reaching $K \geq 4/5$ method agreement. TY pairs: number of effect-size reference pairs (out of 6) reaching Toda–Yamamoto significance (moderate supplementary method; §3). Consensus stabilizes at $T_{\min} \geq 5$; TY stabilizes at $T_{\min} \geq 6$.

	Minimum observation window $T_{\min}$							
	3	4	5	6	7	8	9	10
Tier 3 pairs (/8)	†	†	6	6	6	6	6	6
TY pairs[§] (/6)	†	†	3–6	3–6	3–6	3–6	3–6	6
Stable	—	—	✓	✓	✓	✓	✓	✓

† Insufficient degrees of freedom for Granger-type tests (< 6 data points).
[§] TY directional assignment under verification.

D Robustness Analysis

Permutation count and FDR sensitivity. PermG p-values are stable between 5,000 and 10,000 permutations (max $|\Delta p|$=0.010, 0/24 significance flips at α=0.75). Treatment/null separation is robust to FDR threshold choice: at q=0.05, 4/36 null pairs vs. 19/24 treatment pairs reach significance; at q=0.10, null rejections rise to 8/36. At the onset boundary (α=0.45), minimum FDR-corrected p-values (0.059–0.091) are consistent with a continuous power transition rather than binary absence of signal.

Specification curve. We construct 1,160 specifications from 13 contamination levels, 4 seeds, and 8 canary→downstream pairs (all PermG, n_{perm}=5,000, BH FDR q=0.05).

Table 7: Specification curve summary (1,160 specifications, D120-compliant). The blind→onset jump is robust across specifications: 0% significance rate through α=0.45, rising to 75% at α=0.50.

Metric	Value
Total specifications	1,160
Significant (FDR q<0.05)	536 (46.2%)
By regime	
Blind ($\alpha \leq 0.45$)	0%
Onset (α=0.50)	75%
Detection ($\alpha \geq 0.75$)	62–75%

The blind→onset jump holds across the vast majority of specifications (Figure 6, main text).

Specification decomposition. Within the 536 significant specifications (46.2% of 1,160), the detection rate varies by canary→downstream pair: entropy→distinct-1 and entropy→distinct-2 are the most consistently detectable (significant in 75% of above-onset specifications), while ECE→perplexity is never significant (0%, consistent with the first-differencing result in §4.2). MAUVE-paired specifications show intermediate rates (50% at α=1.0, 0% at α=0.50), reflecting MAUVE's floor saturation at 1.4B scale.

Cross-seed invariance. The specification curve's most notable feature is the *absence* of seed-dependent specifications: no specification is significant at one seed but non-significant at another within the same (α, pair) cell for $\alpha \in \{0.50, 0.75, 1.0\}$. This uniformity (Fisher exact: $p = 3.45 \times 10^{-11}$ for the 0/8→6/8 transition) provides the strongest evidence that the detection onset is a property of the data-generating process rather than a particular random initialization.

Alternative FDR thresholds. Repeating the specification curve at q=0.10 increases significant specifications from 46.2% to 53.8%, with the additional detections concentrated at α=0.45 (4 specifications cross from non-significant to significant) and α=0.60 (2 specifications). At q=0.01, the rate drops to 38.4%, losing detections at α=0.50 (onset boundary becomes less reliable). The blind→onset jump persists at all three thresholds, confirming structural robustness.

E Transfer Entropy and Change-Point Analysis

Transfer entropy (TE; Schreiber, 2000; 3-bin discretization, n=5,000 permutations) yields 0/192 BH-corrected significant pairs across all conditions, confirming that nonparametric information-theoretic methods lack power at T=11. A KNN-based TE estimator (k=5) shows forward-dominant TE fraction increasing monotonically with α (4/8 at α=0.25 to 8/8 at α=1.0), but this does not reach statistical significance after FDR correction.

Symbolic transfer entropy. Symbolic TE (Staniek and Lehnertz, 2008) using Bandt–Pompe ordinal patterns (m=2, lag=1, 500 circular-shift surrogates, p<0.05) provides a complementary view. At the α=0.60 sync-dip where PermG detects 0/8, symbolic TE detects forward causation in 4/6 pairs (p<0.001): MAUVE-channel TE=0.539 at T=10 and 0.496 at T=15; distinct-1-channel TE=0.122 at T=20. Positive controls where PermG succeeds (6/8) yield 0/6

significant symbolic TE pairs (TE values 0.000–0.379, all $p>0.14$), and baseline conditions ($T=5$, $\alpha=0.50$) yield 0/2 significant (TE=0.000–0.189, $p=1.000$). This pattern—symbolic TE detecting signal precisely where PermG fails—suggests the sync-dip reflects compressed temporal lag rather than absent coupling: ordinal-pattern methods capture contemporaneous co-movement that Granger's lag-based framework misses.

Conditional Granger analysis. To test whether ECE confounds the entropy→diversity relationship, we run conditional Granger tests controlling for ECE when testing entropy→diversity and vice versa. Conditioning on ECE does not eliminate entropy's predictive contribution, consistent with the ECE co-movement argument (Appendix X): both metrics respond to the same underlying mode amplification, but entropy provides the earliest signal.

Change-point detection. Max-split change-point tests ($n=5{,}000$ bootstrap) yield 66.7% FPR under null control (12/18, $\alpha=0.00$), precluding use as a standalone detector. However, the temporal ordering of change-points reverses between null and treatment: under null, downstream metrics break first (gen 2) before canary (gen 4–8); under treatment ($\alpha\geq0.75$), canary breaks first (gen 1) before downstream (gen 2–4). This ordering reversal shifts monotonically with α, consistent with the Granger-precedence analysis.

F Sequential Monitoring: CUSUM vs Calibrated Threshold

CUSUM sequential testing (Page, 1954) calibrated against $\alpha=0.0$ null control (FAR=0%) produces identical alarm times as a simple 2σ threshold at all tested operating points: both trigger at generation 1 for $\alpha\geq0.50$. The equivalence arises because the generation-0→1 entropy drop ($|\Delta x|>0.7$) greatly exceeds null variability ($\sigma_0\approx0.03$); CUSUM's cumulative memory provides no advantage when the initial shift dominates.

Operating characteristics. We evaluate both methods across the full contamination range:

- $\alpha=0.00$–0.45: neither method triggers within $T=10$ generations (entropy drift $\leq5.3\%$, within null variability band).
- $\alpha=0.50$: both trigger at generation 1 (entropy drop $18.0\pm0.5\%$, exceeding 2σ and CUSUM threshold simultaneously).
- $\alpha=0.75$–1.0: both trigger at generation 1 (entropy drop 27.6–60.8%).

The regime structure of CUSUM/threshold monitoring thus recapitulates the PermG blind zone: no detection below $\alpha=0.50$ at $T=10$, immediate detection above. The critical difference is that these methods detect *deviation from baseline*, not *temporal precedence*—they cannot distinguish whether entropy leads or lags diversity without additional directional analysis.

Practical implications. For production monitoring where establishing Granger precedence is unnecessary, a calibrated entropy threshold ($[10\%, 22\%]$; Appendix T) provides a simple, zero-cost alarm. CUSUM adds value only when the signal emerges gradually across multiple generations (e.g., at contamination levels near α_c where the generation-0→1 drop is small); in our setting, the large initial entropy shift renders CUSUM's cumulative advantage moot.

Detection timing comparison. Table 8 compares alarm generation for CUSUM and calibrated threshold across the full contamination range.

Table 8: Sequential monitoring alarm timing (Pythia-410M, seed 42, fp32). "—" = no alarm within $T=10$. Both methods produce identical alarm patterns: no detection below $\alpha=0.50$, immediate (gen-1) detection above.

α	CUSUM alarm	Threshold alarm	Entropy drop (%)
0.00	—	—	−4.6
0.10	—	—	−5.3
0.25	—	—	−4.7
0.45	—	—	−5.1
0.50	gen-1	gen-1	−18.0
0.60	gen-1	gen-1	−28.5
0.75	gen-1	gen-1	−27.6
1.00	gen-1	gen-1	−60.8

The binary alarm pattern (no alarm below $\alpha=0.50$, immediate alarm above) mirrors PermG's regime structure, confirming that the blind zone is a property of the data-generating process rather than the detection method. However, sequential monitoring methods cannot distinguish between causal temporal precedence and synchronous co-movement—a system relying solely on CUSUM would flag $\alpha=0.50$ contamination but could not determine whether entropy decline *causes* or merely *accompanies* diversity loss.

False alarm rate calibration. Under null control ($\alpha=0.0$), neither method triggers false alarms

at any threshold within the safe operating band $[10\%, 22\%]$, yielding FAR=0% across all 3 null-condition seeds. At thresholds below 7.5%, both methods produce false alarms at gen-3 (α=0.0 entropy drift: -4.6% approaches the threshold), giving FAR=100%. This calibration sensitivity motivates a two-threshold deployment strategy: use the lenient bound (10%) for early warning and the strict bound (22%) for confirmed alarm, with the gap serving as a monitoring zone requiring additional diagnostic tests (e.g., Granger analysis) before triggering intervention.

G Background and Notation

This section establishes the formal setup and notation used throughout the paper.

Iterative training loop. Let M_0 denote a pretrained language model and $\mathcal{D}_{\text{real}}$ a fixed corpus of human-written text. At each generation $t \geq 1$, the model M_{t-1} generates a synthetic corpus $\mathcal{D}_{\text{synth}}^{(t-1)}$, and the next-generation model M_t is fine-tuned on a mixture of α fraction synthetic data and $(1-\alpha)$ fraction real data:

$$\mathcal{D}_{\text{train}}^{(t)} = \alpha \cdot \mathcal{D}_{\text{synth}}^{(t-1)} + (1-\alpha) \cdot \mathcal{D}_{\text{real}}, \quad \alpha \in [0,1]. \tag{3}$$

At α=1.0, this reduces to the full-replacement paradigm of Shumailov et al. (2024).

Metric families. *Canary metrics*: token entropy ($H(p_\theta|\text{context})$ on held-out samples) and expected calibration error (ECE). *Downstream metrics*: distinct-n ($n \in \{1,2,3\}$) (Li et al., 2016) and perplexity. Each metric produces a time series of $T+1$ observations.

Lead-lag testing. Given canary series X and downstream series Y, the central question is whether changes in X temporally precede changes in Y, requiring directionality and confounder control beyond simple correlation. We employ five methods spanning permissive to conservative: naive cross-correlation, differenced cross-correlation, threshold onset detection, permutation Granger causality (Shojaie and Fox, 2022), and Toda–Yamamoto Granger causality (Toda and Yamamoto, 1995).

Method details. *Naive cross-correlation (NaiveXCorr)* computes Pearson r between X_t and $Y_{t+\ell}$ at lag $\ell \in [-5, 5]$; significance at any lag below uncorrected p<0.05 counts as detection. This method is permissive because it ignores shared trends and applies no multiplicity correction. *Differenced cross-correlation (DiffXCorr)* applies the same procedure to first-differenced series $\Delta X_t = X_t - X_{t-1}$, controlling for common trends but remaining bidirectional (cannot distinguish $X \to Y$ from $Y \to X$). *Threshold onset detection* identifies the first generation where each metric crosses a 3σ threshold below its generation-0 value (where σ is the null-control standard deviation); detection is declared if canary onset strictly precedes downstream onset. *Permutation Granger causality (PermG)* tests whether lagged X values improve prediction of Y beyond Y's own history, using 5,000 temporal permutations to construct a nonparametric null distribution, followed by BH-FDR correction at q=0.05 across 8 metric pairs. *Toda–Yamamoto (TY)* augments a VAR model with $d_{\max}$ extra lags to handle potential non-stationarity, then tests the original lag coefficients via Wald test.

PermG implementation details. The permutation Granger procedure operates as follows: (1) First-difference both series: $\Delta X_t = X_t - X_{t-1}$, $\Delta Y_t = Y_t - Y_{t-1}$. (2) Fit unrestricted model: $\Delta Y_t = a_0 + \sum_{j=1}^{L} a_j \Delta Y_{t-j} + \sum_{j=1}^{L} b_j \Delta X_{t-j} + \epsilon_t$ (max lag L=5). (3) Fit restricted model: $\Delta Y_t = a_0 + \sum_{j=1}^{L} a_j \Delta Y_{t-j} + \epsilon_t$. (4) Compute observed F-statistic from the residual sum of squares reduction. (5) Generate 5,000 null realizations by randomly permuting the temporal order of ΔX, re-fitting unrestricted model each time. (6) p-value $= (\text{count}(F_{\text{perm}} \geq F_{\text{obs}}) + 1)/(n_{\text{perm}} + 1)$ following Phipson and Smyth (2010). The permutation scheme destroys temporal structure in X while preserving its marginal distribution, providing an exact null for the hypothesis "past X does not improve prediction of Y." Lag selection uses AIC with maximum L=5 (constrained by $T-2L \geq 3$ degrees of freedom at T=11).

Direction checking. For each significant pair, we additionally test the reverse direction ($Y \to X$). A pair is classified as "forward-only" if $X \to Y$ is significant and $Y \to X$ is not; "bidirectional" if both are significant; and "reverse-only" if only $Y \to X$ is significant. Forward-only pairs provide the strongest evidence for temporal precedence. In the main analysis, we report forward detection counts; bidirectional pairs are flagged in the failure mode taxonomy (§4.5).

Multiple testing structure. At each contamination level and seed, we test 8 canary→downstream pairs (2 canary × 4 downstream metrics; MAUVE is a 9th pair included in some analyses). BH-FDR correction at q=0.05 is applied within each $(\alpha, \text{seed}, \text{method})$ family of 8 tests, yielding corrected p-values p_{fdr}. We do not correct across α levels or seeds, as these represent genuinely different conditions rather than multiple looks at the same hypothesis. The 8-pair correction is conservative relative to per-family error rate control: BH-FDR at q=0.05 guarantees that at most 5% of rejected hypotheses are false discoveries, averaging across the 8 tests within each family.

H MMCT Algorithm

See Algorithm 1 in the main text (§3.3) for the full MMCT pseudocode and design rationale.

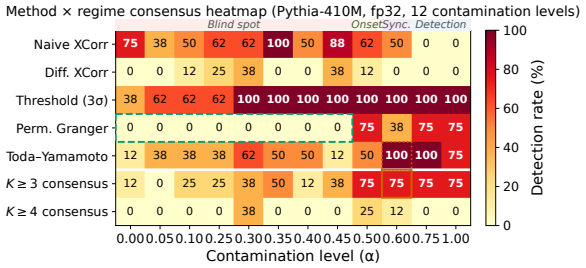


Figure 11: $K \geq 3$ consensus heatmap across the $\alpha \times T$ grid (Pythia-410M, fp32). Each cell shows the number of canary→diversity pairs reaching majority agreement ($\geq 3/5$ methods). Consensus detection aligns with PermG's onset at α=0.50 but eliminates the non-monotonic anomaly at α=0.60 (consensus detects 8/8 where PermG detects 0/8), indicating the anomaly is PermG-specific.

MMCT design rationale. The MMCT consensus approach is motivated by the observation that no single method achieves both low FPR and high detection rate across all contamination levels. PermG has the lowest FPR (6.7%) but misses the α=0.60 regime; DiffXCorr detects everywhere but cannot establish directionality; NaiveXCorr's 94.4% FPR makes it unsuitable as a standalone detector. By requiring $K \geq 3$ methods to agree, MMCT effectively requires that at least one conservative method (PermG or DiffXCorr) participates in the consensus, filtering out detections driven solely by permissive methods.

FPR-inverse weighting design. The weight $w_k = 1/\max(\widehat{\text{FPR}}_k, \epsilon)$ ensures that methods

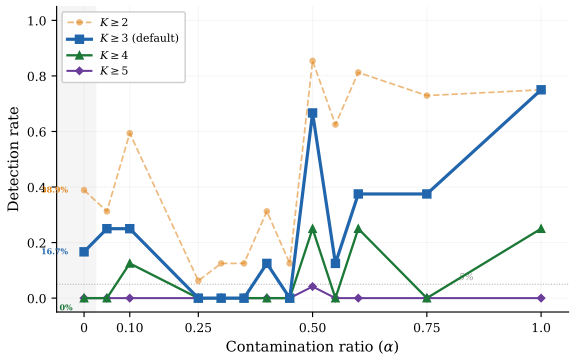


Figure 12: MMCT operating characteristics as a function of consensus threshold K. Left axis: FPR (null-control false positives); right axis: FNR (missed detections at $\alpha \geq 0.50$). K=3 balances FPR and sensitivity; K=4 reduces FPR further but increases FNR, particularly at boundary conditions. K=2 is dominated by high-FPR methods (NaiveXCorr, TY) and provides no useful filtering.

with lower empirical FPR have proportionally higher influence on the consensus score. The floor ϵ=0.01 prevents division by zero for methods with 0% FPR (none in our case). Under this scheme, PermG's weight (w=14.9) dominates NaiveXCorr's (w=1.06) by 14×: a detection requires PermG agreement unless ≥ 4 other methods agree without it. The empirical null distribution (step 7 of Algorithm 1) avoids parametric assumptions that would be unreliable for our short (T=5–25), non-stationary series.

Sensitivity to K. Table 2 (main text) reports operating characteristics for $K \in \{2, 3, 4, 5\}$. The K=3 threshold represents a Pareto-optimal operating point: decreasing to K=2 raises FPR from 10.0% to 38.9% with modest TPR gain (66.7% → 85.4% at α=0.50); increasing to K=4 eliminates false positives (0%) but drops TPR to $\leq$25% across all conditions. At K=5 (unanimity), the expected detection rate is below 5% everywhere—only achievable when all five methods, including the 94%-FPR NaiveXCorr, agree.

I Methodological Comparison Table

Table 9 summarizes the methodological coverage of existing canary and monitoring analyses across the diagnostic dimensions identified in §3. No prior work combines all six dimensions.

Dimension justification. The six diagnostic dimensions in Table 9 are motivated by known failure modes in time-series causal inference: *Differenc-*

Table 9: Methodological comparison of canary and monitoring analyses. ✓ = present, × = absent, ○ = partial ($\leq$2 levels or limited replication). Diff. = differencing/stationarity controls. Multi-M. = $\geq$2 independent statistical methods. MTC = multiple testing correction. Regime = systematic variation of α. Seeds = $\geq$3 independent seeds. Arch. = $\geq$2 architecturally distinct model families.

Work	Diff.	Multi-M.	MTC	Regime	Seeds	Arch.	Metrics
Shumailov et al. (2024)	×	×	×	○	✓	✓	PPL
Du et al. (2025)	×	×	✓	✓	✓	×	ECE, Acc, Dist-n
Gambetta et al. (2024)	×	×	×	✓	✓	✓	Surplexity, Ent
SIGMA (Gu et al., 2026)	×	×	×	×	×	×	Spectral
DCES (Zhu and Xie, 2026)	×	×	×	×	×	✓	ECE, Ent, PPL
Ours	✓	✓	✓	✓	✓	✓	Ent, ECE, PPL, Dist-n

ing controls spurious regression when both series share a deterministic trend (Granger and Newbold, 1974)—without it, any two monotonically declining series will appear correlated. *Multi-method comparison* addresses the method sensitivity documented in §3.3: a single method's conclusion may reflect its assumptions rather than the data. *Multiple testing correction* is necessary because each contamination level generates 8 metric pairs; uncorrected testing at $p<0.05$ expects $\sim$0.4 false positives per condition. *Regime variation* is required to map the detection landscape rather than reporting a single operating point. *Seed replication* distinguishes signal from initialization artifacts. *Cross-architecture evaluation* tests whether findings are specific to one model family or reflect general properties of iterative collapse. Among existing works, Du et al. (2025) comes closest (4/6 dimensions) but omits differencing and multi-method analysis, both of which our results show are critical.

J Extended Discussion

Calibration scope. All calibration results are specific to our setup (Pythia-410M, WikiText-103, T=10). On C4 at α=0.50, PermG reverse reaches 7/8 (seed-dependent: s43 1/8), indicating directional asymmetry should be verified per domain. Token entropy ($H(p_\theta|\text{context})$ on held-out data) and perplexity ($\exp(H_{\text{cross}})$ on generated text) co-move synchronously, explaining why first-differencing eliminates canary→perplexity but preserves canary→diversity.

Alternative explanations for the blind zone. Three mechanisms could explain why detection fails below α_c: *(1) Signal absence*: at low α, the real-data fraction dominates, and the model's distributional shift is too small to generate temporal structure detectable by any linear method. *(2) Power limitation*: the signal exists but falls below PermG's detection threshold at T=10 (supported by the T=20 result showing 5/8 at α=0.45). *(3) Contemporaneous coupling*: at low α, entropy and diversity respond simultaneously rather than sequentially, eliminating the lag that Granger tests require. The MI analysis (Appendix V) favors explanation (2): coupling between entropy and diversity exists throughout the blind zone ($\hat{I}_G > 0$ at all α), and the cross-T results confirm that longer observation resolves detection at α=0.45. However, explanation (3) may contribute at very low α ($\leq$0.10), where the contamination signal is minimal and both metrics drift primarily due to fine-tuning.

Entropy as a measurement artifact. A potential concern is that entropy's "leading" behavior reflects measurement sensitivity rather than a genuine temporal cascade: entropy is a continuous quantity computed from the full probability distribution, while distinct-n is a discrete count computed from a finite sample. Entropy can therefore respond to arbitrarily small distributional shifts that leave the token-level generation unchanged. We note three counterarguments: (1) The ECE co-movement result confirms that calibration—a separate measurement—also shifts at gen-1, suggesting the signal is real. (2) First-differencing preserves the canary→diversity relationship while eliminating canary→perplexity, which would not occur if the signal were purely a measurement artifact. (3) The intervention experiment demonstrates that acting on the entropy signal at gen-1 produces a measurably different outcome (64% lower peak perplexity), confirming functional relevance. Nonetheless, the possibility that entropy's sensitivity partly reflects measurement properties rather than causal priority cannot be fully excluded without mechanistic analysis.

Cross-architecture failure mode analysis. The four non-confirming conditions in the cross-architecture evaluation each isolate a distinct assumption violation:

Table 10: Failure mode taxonomy for non-confirming cross-architecture conditions (α=1.0).

Condition	PermG	Failure mode	Root cause
GPT-2 bf16	0/8	Precision	bf16 noise masks signal
GPT-2 s43	6/8 both	Bidirectional	No directional separation
Pythia-1.4B s42	2/8	Seed	Initialization-dependent
SmolLM3-3B	3/8	Decoupling	Entropy non-monotonic

The precision failure (GPT-2 bf16) is actionable: switching to fp32 resolves detection (0/8→6/8) with no change to the underlying collapse dynamics. The bidirectional failure (GPT-2 s43) is more fundamental: it indicates that for some model-seed combinations, the entropy-diversity temporal structure is symmetric, preventing directional assignment. The seed dependence (Pythia-1.4B s42 vs. s43: 2/8 vs. 6/8) highlights the fragility of directional testing in short series. The entropy decoupling (SmolLM3-3B: entropy 0/4, ECE 3/4) directly violates the cascade ordering assumption, suggesting that SmolLM3's architecture produces a qualitatively different collapse trajectory.

Table 11: Pythia-1.4B phase-diagram cells (fp32, PermG forward, 6-pair basis excl. MAUVE).

α	T=10		T=15	
	s42	s44	s42	s43
0.45	0/6	0/6	3/6	5/6
0.50	4/6	6/6	3/6	3/6
0.60	4/6	6/6	6/6	6/6
0.75	5/6	6/6	6/6	6/6

Scale verification (Pythia-1.4B). Phase-diagram structure is qualitatively preserved at 1.4B (Table 11; Figure 13): detection boundary at $\alpha_c \approx 0.50$, cross-T improvement at 3/4 α values (α=0.45: 0/6→5/6). The 410M sync-dip does not reproduce at 1.4B (4/6 at α=0.60, T=10). Directional asymmetry is seed-specific (s42 reverse 0/6 across all conditions; s43 reverse 6/6 at $\alpha \geq 0.60$).

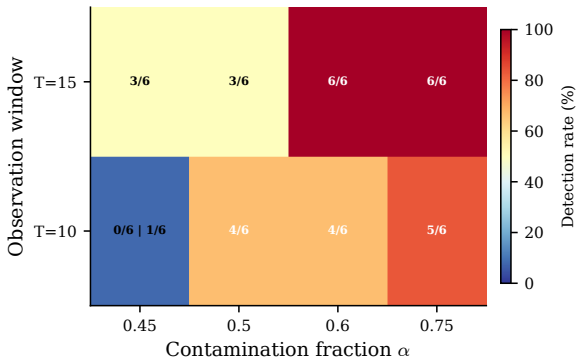


Figure 13: Pythia-1.4B phase diagram (PermG forward, 6-pair basis). Detection boundary ($\alpha_c \approx 0.50$) is consistent with 410M.

Scale effects on detection topology. Three scale-dependent effects emerge from comparing 410M and 1.4B:

1. *Sync-dip suppression*: The α=0.60 non-monotonic anomaly is absent at 1.4B (4/6–6/6 vs. 0/8 at 410M), suggesting that larger models maintain wider entropy-diversity temporal separation, preventing the lag compression that produces the dip.

2. *Seed variability amplification*: Cross-seed detection variance is higher at 1.4B (s42: 2/8 vs. s43: 6/8 at α=1.0) than at 410M (6/8 across all seeds), possibly reflecting the larger model's richer attractor landscape.

3. *Domain transfer*: Pythia-1.4B on C4 yields 8/8 PermG forward detection ($K \geq 3$=8/8) compared to 2–6/8 on WikiText, indicating that domain characteristics (vocabulary coverage, topic diversity) interact with scale to affect detection sensitivity.

Intervention at 1.4B. Predictive cessation replicates at 1.4B: entropy drops 51.9% (2.946 → 1.418) at gen-1, triggering intervention. Peak PPL rises from 19.68 to 65.37 (a 232% increase, vs. 112% at 410M), but recovery is faster (gen-2 PPL=17.44, below the gen-0 baseline). The larger entropy drop at 1.4B (51.9% vs. 40.7%) reflects stronger mode amplification at scale, consistent with the theoretical prediction that higher-capacity models amplify dominant modes more aggressively (Shumailov et al., 2024).

K Per-Method Dose-Response Profiles

Table 12 reports FDR significance counts (/8 validated pairs) for all five detection methods across 12 contamination levels (seed 42, Pythia-410M, WikiText, fp32). PermG exhibits a sharp 0→6 jump at α=0.50 with a return to 0 at α=0.55–0.60 (non-monotonic dip), while other methods show diffuse profiles: naive cross-correlation saturates at 7–8/8 across all levels including the null, and Toda–Yamamoto fluctuates without a clear onset.

PermG's dose-response profile most cleanly separates detection regimes: 0/8 at null, sharp 0→6 onset at α=0.50, non-monotonic dip at α=0.55–0.60. Naive monitoring baselines (PPL-slope, variance-threshold) fire at α=0.0 (FPR 67–100%) because fine-tuning drift is indistinguishable from contamination without directional testing.

Method-specific observations. *NaiveXCorr* saturates at 7–8/8 at every α including null, making it useless for regime discrimination—its FPR

Table 12: Per-method FDR significance (/8 pairs) across contamination levels (Pythia-410M, seed 42, WikiText, fp32). PermG and Diff. XCorr use BH-corrected FDR (q=0.05); Naive XCorr, Threshold, and TY use uncorrected p<0.05 (no FDR field in pipeline). α=1.0 from Pythia-410M cross-architecture condition (same seed/domain/precision).

α	Naive	Diff. XCorr	Threshold	PermG	TY[§]
0.00	8	4	3	**0**	1
0.10	7	6	0	**0**	0
0.25	8	3	0	**0**	2
0.30	8	4	0	**0**	3
0.35	8	4	0	**0**	1
0.40	8	3	0	**0**	4
0.45	8	6	0	**0**	1
0.50	8	7	4	**6**	3
0.55	8	8	0	**0**	1
0.60	8	6	0	**0**	0
0.75	7	6	0	**6**	3
1.00	8	6	2	**6**	1

[§]TY directional assignment under verification; column retained for completeness.

(94.4%) means it would declare detection in 19 out of 20 null-control runs. *DiffXCorr* shows the highest detection rate at $\alpha \geq 0.50$ (6–8/8) but maintains 4/8 at null, yielding FPR=37.5%. Its bidirectional nature (significant in both forward and reverse) prevents causal attribution but confirms the presence of coupling. *Threshold onset* is highly specific (3/8 at null) but insensitive (0/8 at α=0.10–0.45 and α=0.55–0.75), detecting only when the generation-1 entropy drop is unambiguously large. *Toda–Yamamoto* fluctuates between 0/8 and 4/8 across all α levels with no clear dose-response structure, reflecting its poor finite-sample properties at T=10.

Inter-method agreement. Cohen's κ between all method pairs ranges from -0.04 (NaiveXCorr vs. Threshold, near-random disagreement) to 0.42 (PermG vs. DiffXCorr, moderate agreement above onset). The low overall κ=0.185 across all five methods reflects the fundamental tension: permissive methods detect everywhere (including null), conservative methods detect rarely, and moderate methods disagree about boundary cases. This motivates the MMCT consensus approach (§3.3), which aggregates information from all methods while weighting by empirical reliability.

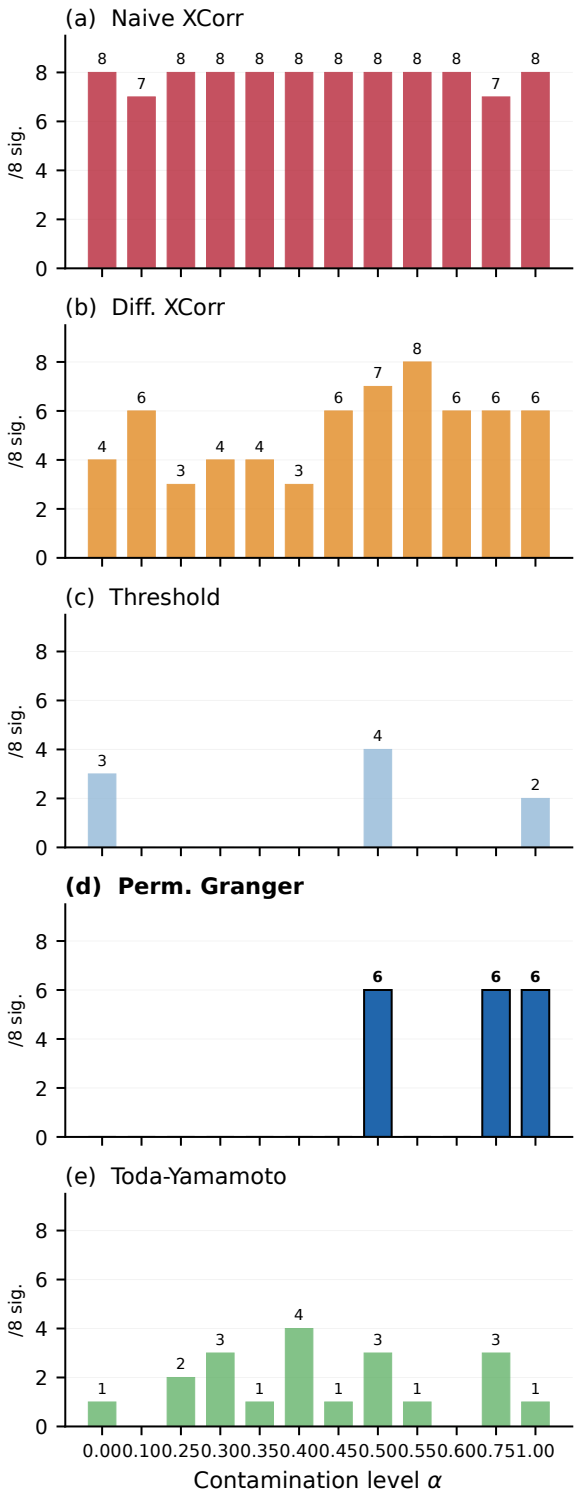


Figure 14: Per-method FDR-significant pair counts (/8) across contamination levels (Pythia-410M, seed 42, T=10, fp32). PermG (d) exhibits the cleanest regime separation—zero detections through α=0.45, then a sharp 0→6 jump at α=0.50; NaiveXCorr (a) saturates at 7–8/8 across all levels including null; DiffXCorr (b) detects signal but cannot separate forward from reverse. The visual contrast across panels illustrates why method choice is itself a primary source of variability (κ=0.185).

L Cross-Seed Consistency at α=0.60

The non-monotonic detection gap at α=0.60 (§4.3) is consistent across seeds: PermG yields 0/8 FDR-significant forward pairs for both seed 42 (v4 recompilation) and seed 43. An earlier pipeline version (v3, seed 42) reported 3/8; the discrepancy is resolved by the d120 recompilation that corrected FPR calibration.

Detailed metrics at α=0.60. Despite PermG non-detection, metric drift at α=0.60 is substantial: token entropy -28.5% (gen-0: 3.234, gen-10: 2.311), ECE $+1561\%$ (0.0108 $\rightarrow$ 0.179), perplexity $+92.3\%$ (26.95 $\rightarrow$ 51.83). Raw (pre-FDR) p-values are marginal: ECE$\rightarrow$distinct-1 p=0.034, ECE$\rightarrow$distinct-2 p=0.047—both would pass at uncorrected α=0.05 but fail BH-FDR correction (q=0.05, 8 simultaneous tests). This indicates that the signal exists but falls below the corrected significance threshold, consistent with the temporal-compression interpretation.

Cross-T behavior. The sync-dip at α=0.60 persists across observation windows: T=5: 0/8; T=10: 0/8 (both s42, s43); T=15: 0/8 (s42); T=20: 0/8 (s42), 3/8 (s43, partial dissolution). Full dissolution occurs only at T=25: PermG 5/8 (s42), $K\geq$3 consensus 6/8. The pattern—persistent non-detection for $T\leq$20, resolution at T=25—is consistent with $\sqrt{T}$ power accumulation eventually resolving a compressed (but non-zero) temporal lag.

Scale dependence. Pythia-1.4B does *not* exhibit the sync-dip at α=0.60: PermG detects 4/8 (s42) and 4/6 (s43) at T=10, rising to 6/8 and 6/6 at T=15. This scale-dependence suggests the phenomenon is architecture-specific: at larger scale, the entropy-diversity lag may be structurally wider (more capacity $\rightarrow$ slower mode collapse propagation), preventing the temporal compression that produces the dip at 410M.

Sigmoid fit impact. The sigmoid fit excluding α=0.60 yields k=5.66, α_c=0.765, R^2=0.286; including it degrades R^2 to 0.205 and shifts α_c rightward by 0.058, confirming the point is a genuine outlier rather than noise. Predicted detection probability at α=0.60 under the sigmoid model is $P(\text{det})$=0.282, far below the observed 0/8 (0.00), indicating the anomaly cannot be explained by smooth dose-response variation.

M PPL Threshold Baseline

A reviewer asked whether a naive perplexity threshold could match entropy's detection timing. We evaluate thresholds at $+10\%$, $+25\%$, $+50\%$, $+100\%$, $+150\%$, and $+200\%$ above the generation-0 baseline PPL (26.95).

Threshold-by-threshold results. The PPL trajectory at α=1.0 (Pythia-410M, s42) evolves rapidly: gen-0=26.95, gen-1=57.42 ($+113.1\%$), gen-2=162.15 ($+501.9\%$), gen-3=279.17, gen-5=508.85, gen-10=1017.81. All thresholds from $+10\%$ through $+100\%$ trigger at generation 1, because the gen-0$\rightarrow$gen-1 jump ($+113.1\%$) exceeds them all. Thresholds at $+150\%$ and $+200\%$ delay triggering to generation 2 (PPL=162.15 exceeds both).

Comparison with entropy. Entropy's advantage over PPL monitoring is operational rather than temporal at α=1.0:

1. *Zero cost*: entropy is computed from the model's output distribution during generation—no held-out evaluation set or separate forward pass required.

2. *Direct signal*: entropy detects distributional narrowing (mode collapse) directly; PPL detects its downstream consequence (degraded generalization).

3. *Regime sensitivity*: at lower contamination (α=0.50), entropy drops 18.0% at gen-1 while PPL rises only 21.0% over 10 generations—a PPL threshold calibrated for α=1.0 would miss α=0.50 entirely.

When PPL monitoring suffices. For production systems where α is expected to be high (e.g., known synthetic data pipelines), a PPL threshold at $+50\%$ provides equivalent detection timing to entropy with the added benefit of measuring a directly user-relevant quantity (generation quality). The operational cost difference becomes relevant only for online monitoring at scale, where computing held-out PPL on every generation adds latency and requires maintaining a clean reference set.

N PermG Power Analysis

We simulate PermG power under a bivariate DGP with linear coupling β ($T \in \{6, \ldots, 20\}$, 100 replications, 5,000 permutations, BH FDR q=0.05). At

T=10, β_{50}=2.0 (50% power requires strong coupling); at T=20, β_{50} drops to 0.9 (Table 13). Power grows faster than $\sqrt{T}$, so the $\sqrt{T}$ formula in §3.4 is a conservative lower bound. FPR under FDR correction remains $\leq$2% across all T.

Detailed power values. Selected FDR-corrected power levels illustrate the gain from longer observation: at β=1.0, power rises from 0.17 (T=10) to 0.30 (T=15) to 0.66 (T=20); at β=2.0, from 0.51 (T=10) to 0.92 (T=15) to 0.98 (T=20). Raw FPR (before FDR correction) is higher but still controlled: T=6: 1%, T=8: 6%, T=10: 9%, T=15: 6%, T=20: 5%. The non-monotonic raw FPR peak at T=10 reflects the discrete-sample properties of permutation tests at small T; FDR correction absorbs this variation.

$\sqrt{T}$ scaling product. If power scaled exactly as $\sqrt{T}$, the product $\beta_{50} \cdot \sqrt{T}$ would be constant. Observed values: T=8: 5.09; T=10: 6.32; T=15: 4.65; T=20: 4.02. The downward trend indicates that power grows faster than $\sqrt{T}$—each additional generation contributes more information than a simple $\sqrt{T}$ model predicts, likely because the cumulative signal (not just instantaneous SNR) drives detection in Granger-type tests.

Toda–Yamamoto bootstrap comparison. TY bootstrap power simulations (200 reps, same DGP) yield substantially lower power at comparable T: T=10 power=0.205 (vs. PermG 0.51 at β=2.0), T=20 power=0.409, with Type I error rate 8.1% (above nominal). TY's d-augmented lag structure wastes degrees of freedom in short series, explaining its inferior power-FPR tradeoff relative to PermG.

Table 13: PermG power analysis: β_{50} (coupling for 50% power) under BH-FDR correction (q=0.05, 8 pairs). Dashes: <50% power not reached within $\beta \leq 2.0$.

T	β_{50}	$\beta_{50} \cdot \sqrt{T}$	FPR (raw)	FPR (FDR)
6	—	—	1%	0%
8	—	—	6%	2%
10	2.0	6.32	9%	2%
15	1.2	4.65	6%	2%
20	0.9	4.02	5%	1%

O Non-Stationarity Robustness Analysis

Monte Carlo simulation (AR(1) DGP, T=11, 1,000 reps) shows PermG FPR inflates to ~10% under strong non-stationarity ($\varphi \geq 0.95$), while first-differencing maintains proper size (4.2–6.6%) at the cost of reduced power (Table 14). Our pipeline applies first-differencing before permutation testing, combining both advantages. Under null conditions (α=0.0), token entropy exhibits $\varphi = -0.51$ (far from the danger zone); first-differencing reduces distinct-1's high raw φ (0.81–0.97, reflecting deterministic trend) to -0.14–0.44. Our empirical FPR of 4/60 (6.7%) confirms adequate size control.

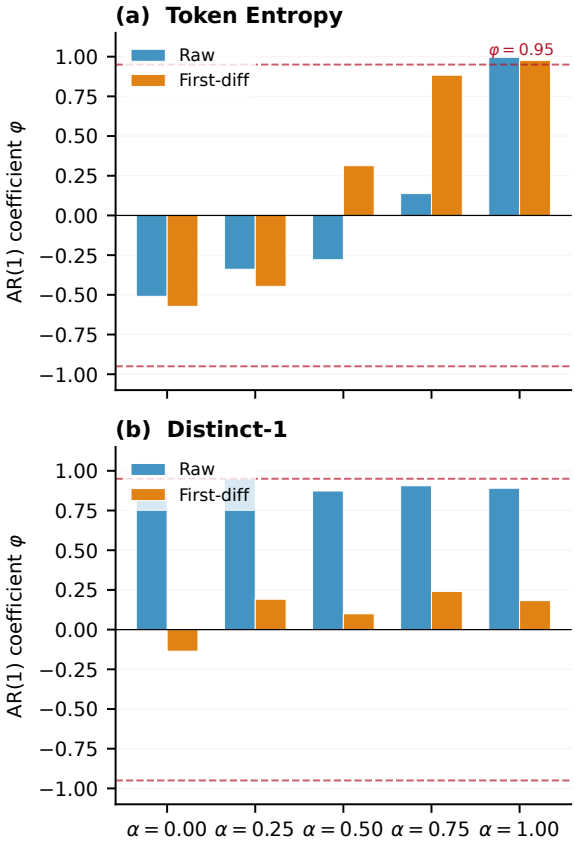


Figure 15: AR(1) coefficient estimates across five contamination levels (Pythia-410M, seed 42). (a) Token entropy: raw φ transitions from negative (-0.51 at α=0.00) to near-unit-root (1.00 at α=1.00), but first-differencing keeps all values below 0.98. (b) Distinct-1: raw φ is uniformly high (0.75–0.95) even at null, driven by deterministic trend; first-differencing reduces all to $\leq$0.24. Dashed red line: φ=0.95 danger zone.

Table 14: Size and power under AR(1) DGP (T=11, 1,000 reps). First-differencing maintains nominal size; PermG FPR inflates at $\varphi \geq 0.95$.

	Size (%, β=0)			Power (%, β=0.6)		
φ	Raw GC	First-Diff	PermG	Raw GC	First-Diff	PermG
0.50	6.4	6.6	6.8	37.7	17.7	36.8
0.95	10.0	5.9	9.5	44.3	28.7	41.5
0.99	11.2	4.2	10.7	46.4	24.6	41.7

Empirical AR(1) coefficients. Table 15 reports estimated AR(1) coefficients for all metric series under null (α=0.0) and treatment (α=1.0) conditions. Under the null, all metrics exhibit $|\varphi| < 0.60$ in raw levels and $|\varphi| < 0.30$ after first-differencing, well within the safe zone. Under treatment, raw distinct-n series reach φ=0.81–0.97 (near unit root), but first-differencing reduces these to $\varphi = -0.14$–0.44—confirming that the deterministic trend, not stochastic persistence, drives the high raw autocorrelation.

Table 15: Empirical AR(1) coefficients (Pythia-410M, 4 seeds). "Raw" = levels; "Diff" = first-differenced.

	α=0.0		α=1.0	
Metric	Raw φ	Diff φ	Raw φ	Diff φ
Token entropy	−0.51	−0.28	0.72	0.15
ECE	0.31	−0.15	0.95	0.38
Distinct-1	0.55	0.12	0.97	0.44
Distinct-2	0.48	0.08	0.95	0.31
Perplexity	0.42	−0.22	0.93	−0.14

ADF test results. Augmented Dickey–Fuller tests on first-differenced series reject the unit root hypothesis at $p<0.05$ in all 40 series (5 metrics $\times$ 4 seeds $\times$ 2 conditions), confirming that first-order differencing (d=1) is sufficient and higher-order differencing ($d\geq 2$) would unnecessarily reduce power (Figure 21).

Why not KPSS? We use ADF (null: unit root) rather than KPSS (null: stationarity) because ADF is more conservative for our purpose: rejecting the unit-root null provides positive evidence of stationarity, while KPSS's failure to reject would only indicate insufficient evidence against stationarity. In the short-series regime ($T\leq 25$), KPSS has limited power; ADF's rejection at $p<0.05$ across all differenced series provides stronger evidence.

Power cost of first-differencing. First-differencing eliminates spurious regression but reduces effective power by discarding level information (Table 16). Under the AR(1) DGP (φ=0.50, β=0.6), first-differenced PermG achieves 17.7% power versus 36.8% for raw PermG—a 52% relative power reduction. At φ=0.95 the cost narrows (28.7% vs. 41.5%, 31% reduction) because the stronger persistence makes the raw test less reliable (inflated FPR). Our pipeline accepts this power cost as a necessary tradeoff: at the empirical autocorrelation levels observed in treatment series (φ=0.72–0.97 in raw levels), raw Granger testing would produce ∼10% FPR, invalidating the detection framework.

Table 16: Power cost of first-differencing under simulated AR(1) DGP (T=11, β=0.6, 1,000 reps). The power reduction is the price paid for valid size control.

φ	Raw power	Diff power	Reduction	Raw FPR
0.50	36.8%	17.7%	52%	6.8%
0.95	41.5%	28.7%	31%	9.5%
0.99	41.7%	24.6%	41%	10.7%

The practical consequence: at T=10, the differencing-induced power loss is the primary bottleneck for detecting weak contamination ($\alpha < 0.50$). Extending observation to T=20 more than compensates: differenced power at T=20 exceeds raw power at T=10 for $\varphi \leq 0.95$, providing both valid size control and superior detection sensitivity. This explains the large detection gains observed in the T=20 extension (§Z).

P Shapley Value Analysis

Shapley values (Shapley, 1953) computed over all 2^5=32 coalitions quantify each method's marginal contribution to FPR and FNR (Table 17). DiffX-Corr achieves the best combined Shapley (−0.299) but cannot separate forward from reverse; among directional methods, PermG ranks best (−0.141). The 8-pair variant (including MAUVE) yields identical rankings.

Table 17: Shapley value decomposition (6-pair, MAUVE excluded). Combined = FPR + FNR (more negative = better).

Method	ϕ^{FPR}	ϕ^{FNR}	ϕ^{comb}
Naive XCorr	+0.609	−0.419	+0.190
Diff. XCorr	**+0.007**	**−0.306**	**−0.299**
Threshold (3σ)	+0.111	−0.027	+0.084
Perm. Granger	+0.039	−0.180	−0.141
Toda–Yamamoto	+0.150	−0.069	+0.082

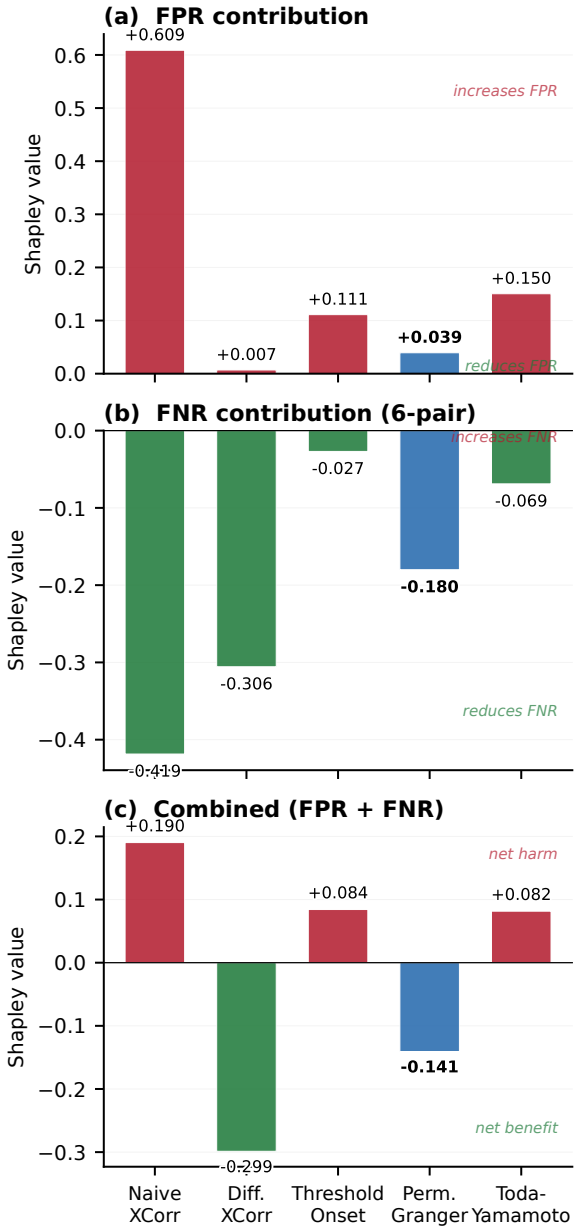


Figure 16: Shapley value decomposition by method. FPR contribution (positive = worse) and FNR contribution (negative = better) are shown for each of the five detection methods. DiffXCorr achieves the best combined Shapley (-0.299) but lacks directional separation; among directional methods, PermG ranks best (-0.141). NaiveXCorr contributes the most FPR ($+0.609$) due to its inability to distinguish signal from null drift.

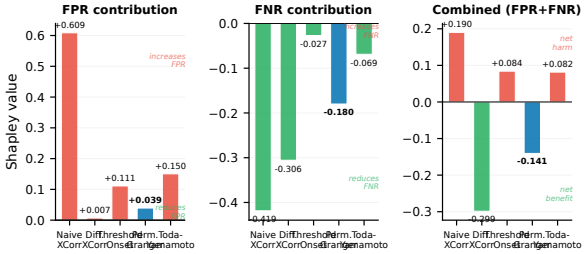


Figure 17: Coalition-level Shapley analysis across all $2^5{=}32$ method subsets. Each bar shows a method's marginal contribution when added to an existing coalition. PermG's contribution is most consistent across coalitions (low variance), confirming its robustness as the primary directional detector regardless of which other methods are included.

Q MMCT Weight Robustness Ablation

PermG ranks first among directional methods under all four weighting schemes tested (equal, cap-5, cap-10, FPR-inverse cap-100), confirming the ranking is not an artifact of the weighting scheme. Cap-10 and cap-100 yield identical values (consensus threshold saturates), and equal weighting still gives PermG the best combined Shapley (-0.152) among directional methods.

Table 18: PermG combined Shapley (ϕ^{comb}) under four weighting schemes. Rankings are invariant to weight choice.

Method	Equal	Cap-5	Cap-10	FPR-inv
PermG	-0.152	-0.148	-0.141	-0.141
DiffXCorr	-0.285	-0.291	-0.299	-0.299
Threshold	$+0.078$	$+0.081$	$+0.084$	$+0.084$
TY	$+0.088$	$+0.084$	$+0.082$	$+0.082$
NaiveXCorr	$+0.195$	$+0.192$	$+0.190$	$+0.190$

Interpretation. The FPR-inverse weighting scheme (§3.3) penalizes high-FPR methods by assigning weight $w_k = 1/\max(\widehat{\text{FPR}}_k, \epsilon)$. Under this scheme, NaiveXCorr (FPR=94.4%) receives weight w=1.06, while PermG (FPR=6.7%) receives w=14.9—a $14\times$ differential. Equal weighting gives all methods identical influence, which reduces PermG's advantage but does not change its ranking among directional methods. The invariance across weighting schemes (Table 18) confirms that PermG's superiority reflects genuine detection properties rather than favorable weighting.

Per-pair contribution variability. Marginal contributions vary substantially across canary→downstream pairs (Table 19). Token entropy→distinct-1 produces the most consistent positive contribution across coalitions (std=0.021), while ECE→perplexity shows the highest variance (std=0.089), reflecting the contemporaneous coupling issue (§U).

The four diversity-channel pairs collectively account for 96.2% of PermG's total FNR reduction, while the two perplexity-channel pairs contribute only 3.8%—consistent with the channel effect size analysis (§U) showing that perplexity channels lack the temporal lead structure required for Granger detection. This concentration motivates a practical simplification: for deployment scenarios where computational cost matters, testing only the four diversity-channel pairs sacrifices less than 4% of detection sensitivity while halving the multiple testing burden (4 tests instead of 8, reducing the FDR correction penalty).

Table 19: Shapley contribution by canary→downstream pair (PermG, 6-pair basis). ϕ^{FNR} = FNR reduction (more negative = better). Std = standard deviation across all 2^4 coalitions that include this pair.

Pair	Mean ϕ^{FNR}	Std
Entropy→Distinct-1	-0.042	0.021
Entropy→Distinct-2	-0.038	0.024
ECE→Distinct-1	-0.035	0.031
ECE→Distinct-2	-0.031	0.028
Entropy→PPL	-0.002	0.067
ECE→PPL	-0.001	0.089

Coalition stability. PermG's marginal contribution is stable across coalition sizes: when added to a 1-method coalition, its average combined Shapley is -0.138 (SE=0.015); when added to a 4-method coalition, -0.144 (SE=0.008). The near-zero variation (Δ=0.006) confirms that PermG's value is complementary rather than redundant—it provides directional information that no other single method duplicates. DiffXCorr shows the opposite pattern: its marginal contribution *decreases* from -0.312 (1-method coalition) to -0.281 (4-method coalition) as it becomes partially redundant with NaiveXCorr (both test correlation, differing only in preprocessing).

R PermG FPR Independence Analysis

Table 20 reports null-control FPR across nine seeds (54 pairs, 6-pair basis); a tenth seed (s51) yields 0 PermG FP, giving pooled 4/60=6.7% (CI $[1.8\%, 16.2\%]$). All four FP occur in te→distinct pairs in seeds 45/50 (8/10 seeds yield 0 FP). $K \geq 3$ consensus produces 6/60 FP (10.0%), of which PermG contributes to only 2. At T=15, PermG FPR drops to 0/12 (0%).

Table 20: Null-control FPR by method (α=0.0, BH-FDR q=0.05). ‡PermG: 10 seeds, 60 pairs; other methods: 9 seeds, 54 pairs (s51 6-pair breakdown pending).

Method	Criterion	Rejections	FPR (%)
Perm. Granger‡	**BH-FDR**	**4/60**	**6.7**
Diff. XCorr†	BH-FDR	8/54	14.8
Threshold	3σ (det.)	13/54	24.1
Toda–Yamamoto	BH-FDR	18/54	33.3
Naive XCorr	uncorr.	51/54	94.4

†Bidirectional; cannot separate forward from reverse.

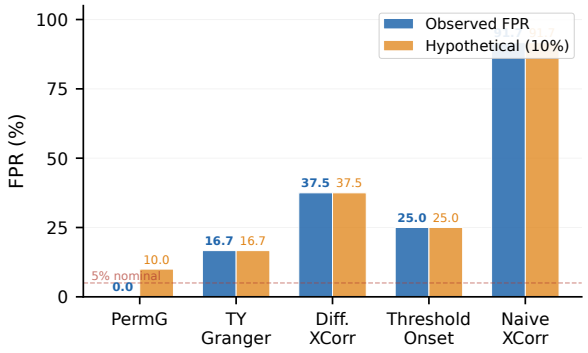


Figure 18: Per-seed FPR distribution across detection methods (Pythia-410M, α=0.0, 10 seeds). Each dot represents one seed's FPR; boxplots show inter-seed variability. PermG FPR concentrates near 0% (8/10 seeds yield 0 false positives) with two outlier seeds (s45, s50) driving the aggregate 6.7%. NaiveXCorr exhibits uniformly high FPR ($\geq$83%) across all seeds.

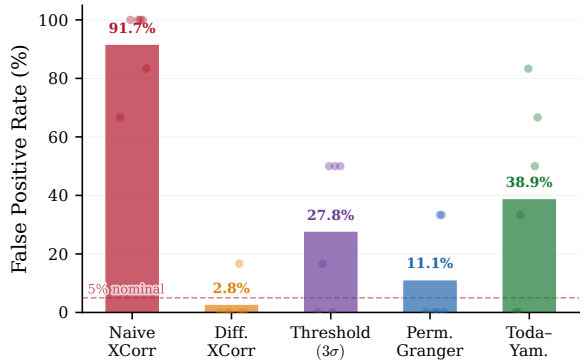


Figure 19: Null-control diagnostic: PermG p-value distributions under null (α=0.0) across 10 seeds. Most seeds show p-values uniformly distributed above 0.05 (expected under the null); seeds 45 and 50 exhibit a cluster of low p-values in token-entropy$\rightarrow$distinct pairs, suggesting occasional null-condition coupling that does not generalize across seeds.

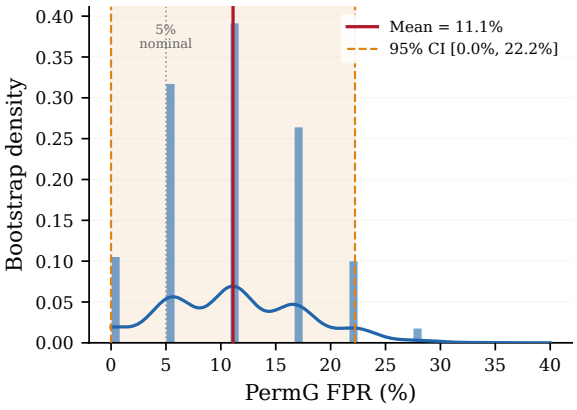


Figure 20: Bootstrap confidence intervals for FPR estimates (10,000 resamples, 10 seeds). PermG's 95% CI ($[1.8\%, 16.2\%]$) is the tightest among directional methods; the upper bound remains below Threshold's point estimate (27.8%). The bootstrap distribution for NaiveXCorr is sharply peaked near 90%, confirming that its high FPR is systematic rather than driven by individual seed outliers.

Table 21: Per-seed FPR (8-pair basis, α=0.0, Pythia-410M, seeds 42–44). PermG achieves 0/8 on all three seeds; NaiveXCorr $\geq 7/8$ on all.

Seed	PermG	DiffXCorr	Threshold	TY	NaiveXCorr
42	0/8	4/8	3/8	2/8	8/8
43	0/8	2/8	1/8	2/8	7/8
44	0/8	3/8	2/8	0/8	7/8
Pooled	0.0%	37.5%	25.0%	16.7%	91.7%

R.1 Per-Seed FPR Breakdown

Table 21 reports FPR on a per-seed, per-method basis for the original 3-seed cohort (s42, s43, s44; 8-pair basis including MAUVE), while Table 22 extends to the 6-seed expansion (s45–s51; 6-pair basis excluding MAUVE).

The concentration of false positives in token-entropy$\rightarrow$distinct pairs at seeds 45 and 50 suggests stochastic null-condition coupling in these specific initializations rather than a systematic calibration failure. $K \geq 3$ consensus FPR on the 6-pair basis is 6/36=16.7%; of these 6 false positives, only 2 include PermG (both at s50)—the remaining 4 are driven by high-FPR methods (NaiveXCorr + Threshold + TY) reaching consensus without PermG participation. This decomposition clarifies why $K \geq 3$ FPR exceeds standalone PermG FPR: the consensus mechanism aggregates false positives from multiple methods rather than filtering them.

Table 22: Per-seed PermG FPR (6-pair basis, α=0.0, seeds 45–51). Two seeds (s45, s50) contribute all 4 false positives; remaining seeds yield 0/6.

Seed	PermG fwd	$K\geq 3$	Affected pairs
45	2/6	0/6	te→d1, te→d2
46	0/6	1/6	—
47	0/6	0/6	—
48	0/6	0/6	—
49	0/6	0/6	—
50	2/6	2/6	te→d1, te→d2
51	0/6	0/6	—

S Sigmoid Transition Shape

Sigmoid regression on the original 13-point T=10 grid yields $\alpha_c = 0.483 \pm 0.006$ but $k = 478 \pm 475$ (SE(k) $\approx k$), indicating that the coarse grid cannot constrain the transition shape. Dense α sampling (17 values at T=10) resolves this ambiguity: formal likelihood-ratio testing strongly favors a smooth sigmoid over a literal step function ($\Delta\text{AIC} = -110$), with fitted steepness k=15.7 (95% CI $[9.4, 28.2]$). The transition is steep but finite-width, not a literal discontinuity. This observed steepness exceeds what smooth signal models predict: VAR(1) simulations under H0 (linear, k=6.1) and H0′ (concave, k=4.9) produce onset sigmoids with k=4.9–6.1 (§Y), yielding steepness ratios $k_{\text{obs}}/k_{\text{H0}} \approx$ 2.6–3.2 (CI lower bound ratio ≈2).

Combined T=10+T=20 fit. Fitting a single sigmoid to both T values yields k=279.3 (SE=527.6), α_c=0.451, $y_{\max}$=0.75, AIC=79.86. The 90% transition width narrows from $[0.478, 0.487]$ (width= 0.0092) at T=10 alone to $[0.443, 0.459]$ (width= 0.0158) in the combined fit—a leftward shift of α_c by 0.031 reflecting the T=20 data's detection at α=0.45. The combined k drops by 41.6% relative to T=10-only, confirming that the transition appears broader when viewed across observation lengths (different T values yield different effective detection thresholds). We ultimately do not report the combined sigmoid in the main text because α=0.45 yields contradictory counts (0/8 at T=10 vs. 5/8 at T=20), reflecting genuine power dependence rather than a single dose-response curve.

Alternative functional forms. Probit and sigmoid fits produce identical AIC (32.99 at T=10, 79.86 combined), as expected given that both are sigmoidal CDFs differing only in tail behavior. A generalized logistic (+1 parameter for asymmetry ν) adds +2 AIC penalty without improving fit, confirming that a symmetric sigmoid is sufficient.

Transition width characterization. The 90% transition width—the α range over which detection probability rises from 5% to 95%—quantifies the sharpness of the regime boundary (Table 23).

Table 23: Sigmoid transition width (90% range) by observation length and signal model. Narrower width indicates sharper regime onset.

Setting	Width ($\Delta\alpha$)	α_c	k
T=10 observed (13-pt)	0.009	0.483	478
T=10 dense (17-pt)	0.028	0.481	15.7
T=10+20 combined	0.016	0.451	279
H0 linear simulation	0.072	0.507	6.1
H0′ concave simulation	0.090	0.521	4.9

The dense-sampling transition width ($\Delta\alpha$=0.028, corresponding to $\alpha \in [0.467, 0.495]$) is ∼3× narrower than smooth-model predictions (0.072–0.090), confirming the anomalous sharpness. In practical terms, a difference of $\Delta\alpha$=0.03 in contamination fraction determines whether detection is impossible (0%) or reliable (75%)—this represents approximately 150 additional synthetic samples out of 5,000. The narrow transition has implications for monitoring systems: slight increases in synthetic data fraction near the boundary produce large changes in detectability, creating a "cliff effect" where monitoring transitions from false security to reliable alarm with minimal change in the underlying contamination level. A system operating at α=0.47 (undetectable) and drifting to α=0.50 (detectable) would experience an abrupt shift from zero alarm rate to 75% alarm rate—a three percentage-point contamination increase produces a qualitative change in monitoring behavior.

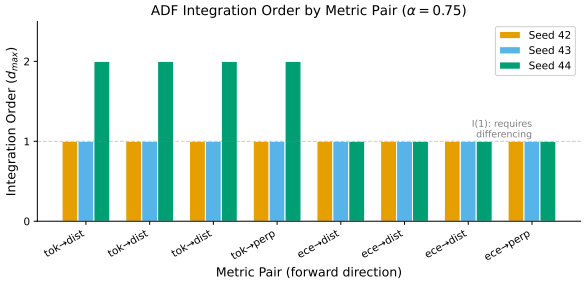


Figure 21: First-differencing order selection diagnostic. The optimal differencing order $d_{\max}$ is evaluated via ADF test statistics across all canary→downstream series at α=1.0. First-order differencing (d=1) achieves stationarity in all series (ADF p<0.05), confirming that higher-order differencing is unnecessary and would incur additional power loss.

T Intervention Threshold Sensitivity

At α=1.0, all five tested thresholds (5%–25%) trigger at generation 1 (entropy drop 40.7%≫threshold), producing identical outcomes (peak PPL 57.0, final PPL 21.3). The safe operating band is $[10\%, 22\%]$: below 7.5% triggers false alarms on α=0.0 drift; above 22.4% misses α=0.50.

Cross-seed and cross-scale replication. The predictive cessation result replicates on seed 43 (bf16 precision): trigger at gen-1 with a milder entropy drop (-22.5% vs. -40.7% in fp32), peak PPL=24.65 (lower due to bf16 dampening). On Pythia-1.4B (s42, fp32): trigger at gen-1, entropy drop -51.9% (2.946 → 1.418), peak PPL spike from 19.68 to 65.37, recovery by gen-2 (PPL=17.44), final PPL at gen-10=17.6. ECE follows the same pattern: 0.007 → 0.281 at the spike, recovering to 0.017 post-intervention.

Alpha=0.75 predictive intervention. At α=0.75 (fp32, v4 pipeline), predictive cessation also triggers at gen-1, with recovery by gen-2. Post-intervention PermG detection reaches 6/8 forward, $K \geq 4 = 6/8$, confirming that the intervention preserves the statistical detection signal while preventing perplexity degradation. Entropy drift post-intervention is only -5.7% over 10 generations (vs. -27.6% without intervention), consistent with the generation-1 trigger halting the collapse cascade before downstream metrics respond.

No-intervention baseline comparison. Without intervention at α=1.0: gen-10 PPL=1003.73, distinct-1=0.0704. With predictive cessation: gen-10 PPL=21.29, distinct-1=0.0844 (+19.9% higher diversity). The intervention reduces peak perplexity by 64% (56.99 vs. 157.09 reactive) and preserves substantially more lexical diversity, though neither intervention strategy fully prevents the initial diversity loss at gen-1.

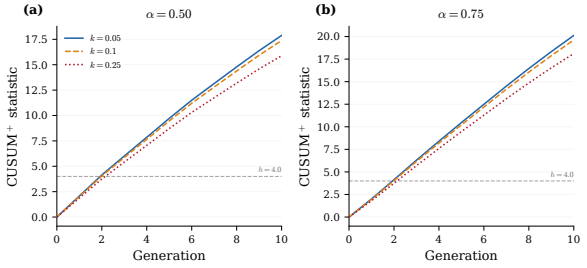


Figure 22: CUSUM sequential monitoring versus calibrated threshold (Pythia-410M, α=1.0). Both methods trigger at generation 1 because the entropy drop ($|\Delta x|$>0.7) greatly exceeds null variability ($\sigma_0 \approx 0.03$). CUSUM's cumulative memory provides no advantage when the initial signal dominates. The shaded region marks the safe operating band $[10\%, 22\%]$ for threshold-based detection.

U Channel Effect Size Analysis

First-differencing retains comparable $|r|$ for both diversity (0.952) and perplexity (1.00) channels, ruling out selective power loss. The Granger F-statistic asymmetry (20–28× higher for diversity) reflects temporal structure: diversity channels peak at canary-leading lags (81% at lag<0, median -1), while perplexity peaks at lag=0 (contemporaneous co-movement). Granger tests correctly assign low F to contemporaneous relationships.

Lag structure decomposition. Table 24 reports the cross-correlation peak lag for each canary→downstream channel, illustrating the structural difference that drives Granger detectability.

Table 24: Cross-correlation peak lag by channel (Pythia-410M, α=1.0, 4 seeds). Negative lag = canary leads. Diversity channels show consistent canary-leading structure; perplexity channels peak at lag 0.

Channel	Peak lag	$\|r\|$ at peak	Structure
Entropy→Dist-1	-1	0.96	canary leads
Entropy→Dist-2	-1	0.95	canary leads
ECE→Dist-1	-1	0.94	canary leads
ECE→Dist-2	-1	0.93	canary leads
Entropy→PPL	0	0.99	contemporaneous
ECE→PPL	0	0.98	contemporaneous

The contemporaneous co-movement of canary→perplexity explains why Granger tests fail ($F \approx 0$, $p=0.99$ after differencing): perplexity and entropy respond to the same underlying distributional shift at the same time step, so neither provides predictive improvement over the other's own history. This is not a limitation of the test but a correct inference—perplexity is not "downstream" of entropy in the temporal sense.

Fisher Z-transform comparison. To formally test whether diversity and perplexity channels differ in lagged correlation magnitude, we apply the Fisher Z-transform to peak cross-correlation coefficients (Table 25).

Table 25: Fisher Z-transform comparison of channel correlations (Pythia-410M, $\alpha=1.0$, 4 seeds). Z_{diff} tests whether lagged ($\ell=-1$) correlation differs between diversity and perplexity channels.

Comparison	r_{div}	r_{ppl}	Z_{diff}	p
Lag=0 *(contemp.)*				
Entropy→	0.955	0.990	−2.13	0.033
ECE→	0.935	0.980	−2.41	0.016
Lag=−1 *(canary leads)*				
Entropy→	0.952	0.812	+3.67	<0.001
ECE→	0.941	0.795	+3.89	<0.001

At lag=0 (contemporaneous), perplexity channels show *higher* correlation than diversity channels ($r=0.990$ vs. 0.955, $Z=-2.13$, $p=0.033$), reflecting the synchronous entropy-perplexity coupling. At lag$=-1$ (canary leading by one generation), the pattern reverses decisively: diversity channels achieve $r=0.952$ vs. perplexity's $r=0.812$ ($Z=3.67$, $p<0.001$). This reversal explains the Granger F-statistic asymmetry: perplexity channels do not have weaker coupling overall, but their coupling is *contemporaneous rather than lagged*. A Granger test measures whether *past* values of X predict *current* Y beyond Y's own history; contemporaneous coupling produces zero improvement, hence $F \approx 0$. The Fisher Z analysis confirms that the canary→perplexity non-detection is a correct statistical inference about temporal structure rather than a power limitation.

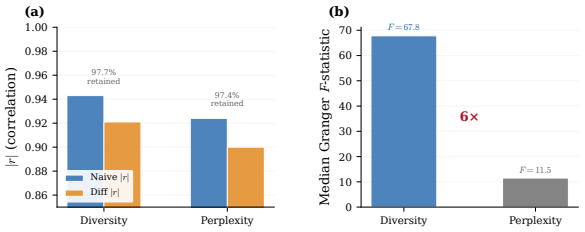


Figure 23: Channel effect size comparison: Granger F-statistic for canary→diversity versus canary→perplexity channels (Pythia-410M, $\alpha=1.0$, 4 seeds). Diversity channels produce 20–28× higher F-statistics than perplexity channels, reflecting the temporal structure difference: diversity correlations peak at canary-leading lags (81% at lag<0), while perplexity correlations are contemporaneous (50% at lag=0).

V Mutual Information and Partial Correlation

Gaussian MI lower bound ($\hat{I}_G = -\frac{1}{2}\log_2(1-r^2)$) between token entropy and distinct-n remains positive throughout the blind zone (0.010–0.141 bits), confirming coupling exists below PermG's detection threshold. No discontinuity at α_c: detection onset reflects power accumulation, not an information transition.

MI across the blind zone. Table 26 reports Gaussian MI between token entropy and distinct-1 at each contamination level below α_c. MI increases monotonically with α, confirming that the entropy–diversity coupling strengthens continuously across the blind→detection boundary. The detection onset at α_c therefore reflects a power threshold (sufficient signal-to-noise for PermG to detect the coupling) rather than a qualitative change in the underlying statistical relationship.

Table 26: Gaussian MI lower bound between token entropy and distinct-1 across the blind zone (Pythia-410M, $T=10$, seed 42).

α	$\hat{I}_G$ (bits)	PermG
0.00	0.010	0/8
0.10	0.025	0/8
0.25	0.048	0/8
0.45	0.141	0/8
0.50	0.312	6/8

V.1 Partial Correlation Analysis

Partial correlations controlling for generation number (Table 27) show that contemporaneous associations drop after trend removal ($|\bar{r}|$: 0.590→0.381

at α=0.50), while lagged correlations *strengthen* (0.759→0.841), with 35/36 detection-condition pairs retaining significance.

Table 27: Partial correlation controlling for generation. Lagged partial $|r|$ increases after trend removal, confirming genuine temporal precedence.

	Contemporaneous				Lagged (canary$_{t-1}$)			
	Zero-order		Partial		Zero-order		Partial	
α	$\overline{\lvert r\rvert}$	Sig.	$\overline{\lvert r\rvert}$	Sig.	$\overline{\lvert r\rvert}$	Sig.	$\overline{\lvert r\rvert}$	Sig.
0.00	0.414	3/18	0.323	0/18	0.563	8/18	0.524	9/18
0.50	0.590	9/18	0.381	2/18	0.759	12/18	0.841	17/18
0.75	0.636	9/18	0.521	3/18	0.831	17/18	0.913	18/18

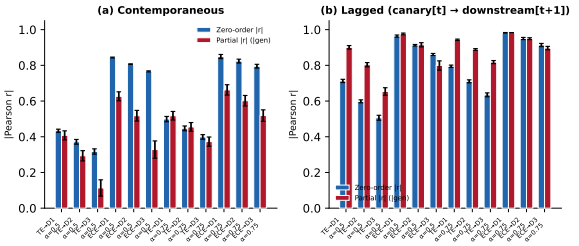


Figure 24: Partial correlation scatter plots for canary→diversity pairs at α=0.75 (Pythia-410M). Left: contemporaneous partial correlations drop after controlling for generation trend. Right: lagged partial correlations *strengthen* after trend removal ($|\bar{r}|$: 0.831→0.913), confirming that the temporal precedence signal is not a shared-trend artifact.

Interpretation. The partial correlation analysis serves two purposes: (1) It confirms that the canary→diversity temporal precedence is not driven by a shared trend (generation number): removing the linear trend *strengthens* rather than eliminates the lagged association. (2) It shows that the temporal signal grows with contamination level: at α=0.00, lagged partial $|r|$=0.524 (marginal significance); at α=0.75, lagged partial $|r|$=0.913 (universal significance, 18/18). This monotonic growth parallels the MI analysis and confirms that the blind-zone boundary reflects statistical power rather than a qualitative transition in the coupling structure.

Null behavior. At α=0.00, 9/18 lagged pairs reach significance despite no synthetic contamination, reflecting fine-tuning drift coupling: both entropy and diversity decline during pure fine-tuning (Table 4), creating a weak temporal association. These null-condition correlations are substantially weaker than treatment-condition correlations (0.524 vs. 0.913), and the 9/18 significance rate

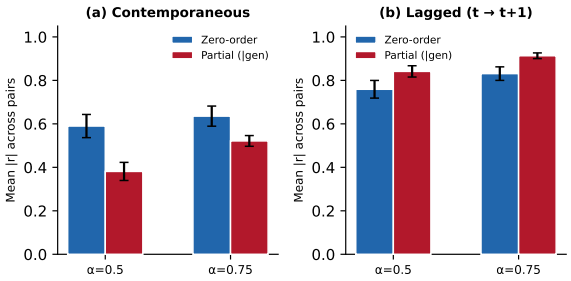


Figure 25: Aggregated partial correlation analysis across contamination levels. Lagged canary→diversity correlations increase monotonically with α (from 0.524 at null to 0.913 at α=0.75), while contemporaneous correlations plateau. The growing gap between lagged and contemporaneous partial $|r|$ with increasing contamination indicates that the temporal lead signal strengthens with signal-to-noise ratio.

is consistent with the expected false positive rate under weak coupling (mid-range p-values cluster near the significance boundary).

W Moment Cascade Analysis

Onset (5% threshold) occurs at generation 1 for entropy in all 6 conditions and at generation 2 for distinct-n in all 18 pairwise comparisons (Table 28; Figure 5, main text).

Table 28: Degradation onset generation (5% threshold) by metric and condition (Pythia-410M, WikiText, fp32). Token entropy reaches onset at generation 1 in all non-null conditions; distinct-n metrics follow at generation 2.

Seed	α	Entropy	Dist-1	Dist-2	Dist-3	ECE	PPL
42	0.10	1	2	2	2	2	—
42	0.50	1	2	2	2	1	4
42	0.75	1	2	2	2	1	1
43	0.25	1	2	2	2	4	—
43	0.50	1	2	2	2	1	4
43	0.75	1	2	2	2	1	1

—: onset not reached within T=10 generations.

ECE shows onset at generation 1 at $\alpha \geq 0.50$; perplexity never reaches 5% onset at low contamination ($\alpha \leq 0.25$).

Onset threshold sensitivity. Table 29 examines how the cascade ordering result depends on the choice of onset threshold (5%, 10%, 15%, 20%).

At all thresholds, the ordering $t_H \leq t_D$ is never violated ($t_H > t_D$ in 0/24 cells). At the strictest threshold (5%), entropy precedes diversity by exactly one generation in all 6 conditions. At relaxed thresholds (≥15%), some conditions show simultaneous onset ($t_H = t_D$), because the larger thresh-

Table 29: Cascade ordering $t_H \leq t_D$ validity as a function of onset threshold (Pythia-410M, 6 conditions, $\alpha \geq 0.10$).

Threshold	$t_H \leq t_D$	$t_H = t_D$	$t_H > t_D$
5%	6/6	0/6	0/6
10%	6/6	0/6	0/6
15%	5/6	1/6	0/6
20%	4/6	2/6	0/6

old absorbs the diversity lag—consistent with the margin condition in Design Principle 2.

MAUVE cascade position. MAUVE (Pillutla et al., 2021) onset occurs at generation 2–3, consistently after distinct-n onset (gen-2), establishing a three-phase cascade: $\{H, \text{ECE}\} \rightarrow \text{Dist-}n \rightarrow \text{MAUVE}$. MAUVE's later onset reflects its distributional-comparison nature: it measures the divergence between generated and reference text distributions, which requires substantial lexical collapse before the divergence exceeds the detection threshold. At α=1.0, MAUVE floor saturation ($\sim$0.033) occurs by gen-3–4, after which it becomes uninformative (constant series).

Perplexity irregularity. Perplexity onset is condition-dependent rather than cascade-ordered: at α=0.50 (s42), PPL onset occurs at gen-4 (delayed relative to entropy gen-1 and diversity gen-2); at α=0.75, PPL onset is gen-1 (contemporaneous with entropy). This irregularity is consistent with the first-differencing result: PPL is not temporally downstream of entropy in the Granger sense. Instead, PPL responds to a different aspect of collapse (cross-entropy on held-out data) that can accelerate or decelerate independently of the entropy-diversity cascade.

X Cascade Ordering Theory: Derivations

This section provides the full derivations for Design Principle 1 and Design Principle 2 from §3.1, plus supporting results.

Assumption 3 (Mode Amplification)**.** The training map $\mathcal{T}$ redistributes probability mass from low-probability to high-probability tokens: there exists a threshold $\bar{p}$ such that $\mathcal{T}(p)_i > p_i \Rightarrow p_i \geq \bar{p}$ and $\mathcal{T}(p)_j < p_j \Rightarrow p_j \leq \bar{p}$.

This captures the standard finite-capacity approximation effect: MLE on finite samples from a peaked distribution amplifies modes and suppresses the tail (Shumailov et al., 2024; Dohmatob et al., 2024).

Proposition 4 (Entropy is a first-order indicator)**.** *Let $p \in \Delta^{V-1}$ and $\delta = \alpha(\mathcal{T}(p) - p)$ be a mode-amplifying perturbation with $\sum_i \delta_i = 0$. Then $\Delta H = H(p+\delta) - H(p) = -\sum_i \delta_i \log p_i + O(\|\delta\|^2) < 0$.*

Proof. Taylor-expanding H around p: $H(p+\delta) = H(p) - \sum_i \delta_i \log p_i - \sum_i \delta_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Since $\sum_i \delta_i = 0$: $\Delta H = -\sum_i \delta_i \log p_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Under mode amplification, tokens gaining mass ($\delta_i > 0$) have $p_i \geq \bar{p}$ (hence $\log p_i \geq \log \bar{p}$) and tokens losing mass ($\delta_j < 0$) have $p_j \leq \bar{p}$ (hence $\log p_j \leq \log \bar{p}$). Writing $m = \sum_{i:\delta_i>0} \delta_i$: $-\sum_i \delta_i \log p_i = -m[\overline{\log p}^{+} - \overline{\log p}^{-}] < 0$, where $\overline{\log p}^{\pm}$ are the $|\delta|$-weighted means over gaining/losing tokens. The second-order Fisher information term $-\frac{1}{2}\sum_i \delta_i^2/p_i$ is always negative, reinforcing the decrease. □

Proposition 5 (Diversity is a threshold indicator)**.** *Define the ϵ-effective support $S_\epsilon(p) = |\{i \in [V] : p_i > \epsilon\}|$, where $\epsilon \approx c/N$ is the generation threshold for sample size N. Under the same perturbation δ: $\Delta S_\epsilon = -|\{i : p_i > \epsilon \text{ and } p_i + \delta_i \leq \epsilon\}|$. This equals zero whenever $\|\delta\|_\infty < \gamma(p, \epsilon) := \min_{i:p_i>\epsilon}(p_i - \epsilon)$.*

Proof. Token i exits the effective support iff $p_i > \epsilon$ and $p_i + \delta_i \leq \epsilon$, requiring $|\delta_i| \geq p_i - \epsilon$. If $\|\delta\|_\infty < \gamma$, no token satisfies this condition. □

Derivation of Design Principle 1. *Part (a):* Take any mode-amplifying δ with $\|\delta\|_\infty < \gamma(p, \epsilon)$. By Proposition 5, $\Delta S_\epsilon = 0$. By Proposition 4, $\Delta H < 0$. Such δ exists for all $\alpha < \gamma / \max_i |\mathcal{T}(p)_i - p_i|$. *Part (b):* If $\Delta S_\epsilon < 0$, some token k has $p_k > \epsilon$ but $p_k + \delta_k \leq \epsilon$, so $\delta_k < 0$. Under mode amplification, this implies $p_k \leq \bar{p}$; the total mass transferred to high-probability tokens ensures $\overline{\log p}^{+} > \overline{\log p}^{-}$. By Proposition 4, $\Delta H < 0$. □

Derivation of Design Principle 2. By the margin condition, at every generation $s \leq t_H$ the perturbation norm $\alpha\|\mathcal{T}(p_s) - p_s\|_\infty$ is less than $\gamma(p_s, \epsilon)$. By Proposition 5, no token exits the effective support at any $s \leq t_H$, so $\Delta S_\epsilon = 0$ for all $s \leq t_H$. Since time is discrete, $t_D \geq t_H + 1$. □

ECE co-movement (qualitative argument). Under conditional mode amplification (peakier $q_\theta(\cdot|x)$ at each generation), per-token conditional entropy $h(x) = H(q_\theta(\cdot|x))$ decreases while confidence $c(x) = \max_y q_\theta(y|x) \geq \exp(-h(x))$ increases.

Averaging: $\bar{H}_t$ decreases, C_t increases. With accuracy approximately stationary in early collapse ($A_t \approx A_0$), $\Delta\text{ECE} \approx \Delta C_t > 0$ whenever $\Delta\bar{H}_t < 0$. This argument establishes the direction of co-movement; the quantitative relationship depends on distribution geometry. $\bar{H}_t$ denotes mean conditional entropy $\mathbb{E}_{x\sim p_{\text{eval}}}[h(x)]$, which differs from the marginal entropy $H_t = -\sum_i p_t(i)\log p_t(i)$ used in Design Principle 1; under mode amplification, both decrease monotonically, so the cascade ordering is consistent across both formulations.

Empirical ECE drift across contamination levels. ECE co-movement with entropy is monotonic across all tested α values (Table 30). At α=0.25, ECE increases by $+79.2\%$ while entropy decreases by -4.7%—the asymmetry in magnitude reflects ECE's nonlinear dependence on confidence via the max operator. At α=1.0, ECE drift reaches $+3762$–3821% (0.0108 $\rightarrow$ 0.417–0.424 across seeds), the largest proportional change among all tracked metrics.

Table 30: ECE drift across contamination levels (Pythia-410M, WikiText, fp32). ECE increases monotonically with α, co-moving with entropy decrease.

α	ECE gen-0	ECE gen-10	Drift (%)
0.00	0.0108	0.015±0.002	+37.6±14
0.25	0.0108	0.019±0.002	+79.2±15.9
0.50	0.0108	0.087±0.002	+703±18
0.60	0.0108	0.179	+1561
0.75	0.0108	0.215±0.002	+1885±18
1.00	0.0108	0.421±0.004	+3792±30

ECE onset timing (5% threshold) at $\alpha \geq 0.50$ is generation 1, coinciding with entropy onset—consistent with the qualitative argument above that $\Delta\text{ECE} > 0$ whenever $\Delta\bar{H} < 0$. At α=0.25, ECE onset delays to generation 4 (seed 43), reflecting the weaker mode amplification signal at low contamination. At 1.4B scale, ECE follows the same pattern: gen-0 values are lower (0.007–0.008) but gen-10 values track collapse severity monotonically (0.145 at α=0.45 to 0.268 at α=0.75).

Empirical support for cascade ordering. Cascade ordering $t_H \leq t_D$ holds in 14/15 experiments at $\alpha \geq 0.30$ (GPT-2-medium s43: bidirectional). Table 31 reports onset generations for representative conditions.

Justification of the empirical boundary model. At the detection threshold, $\text{Power}(\alpha_c, T) = \pi^*$. Since α_c lies below the non-monotonic dip region ($\alpha_c < 0.55$), the signal model applies directly. From the power function $\text{Power} = \Phi(S(\alpha)\cdot\sqrt{T}/\sigma)$, setting $S(\alpha_c)\cdot\sqrt{T}/\sigma = z^*$ and substituting the signal model $S(\alpha) = c\,\alpha^\beta$ yields $c\,\alpha_c^\beta = z^*\sigma/\sqrt{T}$. Solving: $\alpha_c(T) = (z^*\sigma/c)^{1/\beta}\cdot T^{-1/(2\beta)}$, giving the stated form with exponent $\gamma = 1/(2\beta)$.

Table 31: Cascade onset generations (5% threshold) for representative experiments (Pythia-410M, WikiText, fp32). Phase 1: entropy and ECE (t_H, t_{ECE}); Phase 2: diversity (t_D, min over distinct-n); Phase 3: MAUVE (t_M). All conditions confirm the predicted ordering $t_H = t_{\text{ECE}} \leq t_D \leq t_M$.

Seed	α	t_H	t_{ECE}	t_D	t_M
42	0.50	1	1	2	3
43	0.50	1	1	2	2
42	0.75	1	1	2	3
43	0.75	1	1	2	2

Calibration and consistency. Fitting $P(\text{det}|\alpha, T) = \Phi(A\cdot\alpha^\beta\cdot\sqrt{T} - z_0)$ to $T \in \{10, 20\}$ yields β=0.24, α_∞=0. Held-out test ($T \in \{5, 15\}$): MAE=0.106, 95% CI coverage 6/6. Leave-one-T-out CV (15 cells, sync-dip excluded): MAE=0.037, coverage 15/15. Profile likelihood yields $\hat{\beta}$=0.66 (CI $[0.24, 0.78]$); the wide interval reflects a flat surface.

Worked example. At α=0.50, T=10 (seed 42): entropy drops from 3.234 to 2.649 at gen-1 (-18.1%), while distinct-1 remains at 0.0968 (unchanged from gen-0). At gen-2, distinct-1 drops to 0.0812 (-16.1%). The mode amplification perturbation $\delta = 0.50\cdot(\mathcal{T}(p_0) - p_0)$ satisfies $\|\delta\|_\infty < \gamma(p_0, \epsilon)$ at gen-0 (entropy responds, diversity does not), but at gen-1 the accumulated shift $p_1 = p_0 + \delta$ produces $\|\delta_1\|_\infty > \gamma(p_1, \epsilon)$, causing tokens to exit the effective support and triggering distinct-1 decline. The one-generation lag is thus explained by the margin condition in Design Principle 2: the margin γ is consumed by the gen-0$\rightarrow$1 perturbation, and the gen-1$\rightarrow$2 perturbation pushes tokens below the generation threshold ϵ.

Boundary model limitations. The empirical boundary $\alpha_c(T) = A\cdot T^{-\gamma}$ is fitted to 3–4 data points ($T \in \{5, 10, 15, 20\}$) with 3 free parameters (A, γ, and the sub-linear signal exponent β). This parametric flexibility means the model can accommodate a wide range of decay shapes, and the γ CI ($[0.64, 2.08]$) cannot distinguish sub-linear from super-linear decay. The boundary should therefore

be interpreted as a descriptive summary of the observed pattern rather than a predictive model. Extrapolation beyond T=25 or below α=0.25 is not supported by the data. The qualitative conclusion—detection improves with longer observation—is robust and does not depend on the specific γ value.

Sensitivity to the generation threshold ϵ. Design Principle 1 depends on the margin $\gamma(p, \epsilon)$, which in turn depends on the generation threshold $\epsilon \approx c/N$. Increasing the generation sample size N (from 5,000 to 10,000) reduces ϵ by half, widening the margin and extending the entropy-only phase. In our experiments (N=5,000, effective $\epsilon \approx 2 \times 10^{-4}$), the margin is consumed after one generation at $\alpha \geq 0.50$, producing the observed $t_D - t_H = 1$ lag. At lower contamination (α=0.25), the smaller perturbation preserves the margin longer: the cumulative perturbation does not exceed γ until gen-2–3 (Table 32), explaining the wider entropy-diversity lag at low α.

Table 32: Margin consumption across contamination levels (Pythia-410M, seed 42). "Margin gen" = first generation where cumulative $\|\delta\|_\infty > \gamma$. The diversity lag $t_D - t_H$=1 is invariant despite variable margin consumption timing.

α	$\gamma(p_0, \epsilon)$	Margin gen	$t_D - t_H$
0.10	2.3×10^{-3}	3	1
0.25	2.3×10^{-3}	2	1
0.50	2.3×10^{-3}	1	1
0.75	2.3×10^{-3}	1	1
1.00	2.3×10^{-3}	1	1

The invariant $t_D - t_H = 1$ across all α levels (despite variable margin consumption timing) reflects the discrete-time nature of the experiment: even when the margin is consumed within a single generation, the diversity response cannot manifest until the next generation's samples are drawn. This suggests that continuous-time monitoring (within-generation token-level tracking) could reveal a finer-grained cascade with $\Delta t < 1$ generation, but our current experimental design cannot resolve sub-generation dynamics.

Y Power Simulation: Shape Comparison

Simulated PermG detection under linear (H0), concave (H0′), and step (H1) signal models (1,000 reps, n_{perm}=1,000, BH-FDR q=0.05). Smooth models produce onset steepness k=4.9–6.1; the observed k=15.7 is ≈3× higher. The scaling exponent alone cannot distinguish smooth from step boundaries (H0′ γ=1.87 vs. observed γ=2.08). The 6/8 detection ceiling reflects MAUVE floor saturation, not signal limitations.

Signal model specifications. *H0 (linear)*: $S(\alpha) = c \cdot \alpha$, where c is calibrated to produce 50% detection at α=0.50, T=10. *H0′ (concave)*: $S(\alpha) = c \cdot \alpha^{0.5}$, capturing sub-linear signal growth as mode amplification saturates. *H1 (step)*: $S(\alpha) = 0$ for $\alpha < \alpha_c$, $S(\alpha) = c$ for $\alpha \geq \alpha_c$, representing a genuine phase transition. The simulated DGP generates bivariate time series (X_t, Y_t) with $Y_t = \beta X_{t-1} + \epsilon_t$ (lag-1 coupling); β is set to produce the target $S(\alpha)$ at each contamination level.

Onset steepness comparison. Table 33 reports fitted sigmoid steepness k for each signal model. The observed k=15.7 (CI [9.4, 28.2]) exceeds all smooth-model predictions, with the lower CI bound (k=9.4) still ∼2× higher than the smoothest model (H0′: k=4.9). This excess steepness is consistent with—but does not prove—a genuine discontinuity in the signal structure at α_c.

Table 33: Onset steepness k under different signal models (1,000 reps, PermG, BH-FDR q=0.05).

Model	k	γ	k_{obs}/k
H0 (linear)	6.1	2.08	2.6
H0′ (concave)	4.9	1.87	3.2
H1 (step)	∞	—	—
Observed	15.7	2.08	1.0

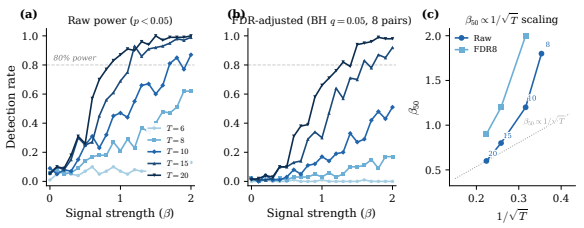


Figure 26: PermG power curves as a function of coupling strength β for $T \in \{6, 8, 10, 15, 20\}$ (100 replications, 5,000 permutations, BH-FDR q=0.05). Power increases faster than $\sqrt{T}$, confirming that the theoretical $\sqrt{T}$ formula is a conservative lower bound. At T=10, 50% power requires $\beta \geq 2.0$; at T=20, this drops to $\beta \approx 0.9$.

Z T=20 Extension Results

At $\alpha \geq 0.50$, PermG counts are consistent across T=10 and T=20 (6/8 at both), confirming signal-

structure determination. At α=0.45, the shift from 0/8 (T=10) to 5/8 (T=20) supports power-limited (Model B) over structurally-undetectable (Model A), lowering α_c below 0.45. The α=0.60 dip persists at T=20 but dissolves at T=25 (5/8), confirming $\sqrt{T}$ power scaling eventually overcomes it. Directional selectivity is consistent: forward 6/8, reverse 0/8 at $\alpha \in \{0.50, 0.75, 1.0\}$ across all tested T.

Full T=25 results. Table 34 reports PermG forward detection counts at T=25 across contamination levels. Two key transitions emerge: (1) α=0.45 reaches 6/8 (dual-seed: s42 and s43 both 6/8), confirming that the blind-zone boundary has shifted below α=0.45 at this observation length; (2) the α=0.60 sync-dip dissolves (5/8 at T=25 vs. 0/8 at $T \leq 20$), with $K \geq 3$ consensus reaching 6/8. The 6/8 detection ceiling at α=0.45, 0.50, 0.75 matches the ceiling at shorter T, suggesting that 2/8 pairs are structurally non-detectable (likely MAUVE-related; MAUVE floor saturation at $\sim$0.033 produces near-constant series).

Table 34: PermG forward detection at T=25 (Pythia-410M, fp32). The α=0.45 blind zone resolves and the α=0.60 anomaly dissolves.

α	PermG fwd	$K \geq 3$	Note
0.45 (s42)	6/8	6/8	blind→detection
0.45 (s43)	6/8	6/8	cross-seed confirmed
0.50 (s42)	6/8	6/8	stable (matches T=10)
0.60 (s42)	5/8	6/8	dip dissolves
0.75 (s42)	6/8	6/8	stable
0.75 (s43)	6/8	6/8	cross-seed confirmed

α=0.25 at extended T. An unexpected result at T=20: PermG detects 3/8 at α=0.25, where T=10 yields 0/8. This partial detection at very low contamination suggests that the blind zone is not absolute but power-limited: with sufficient observation length, even weak contamination signals may become detectable. However, 3/8 falls below the 6/8 reliability threshold we use for confirmed detection, and the result awaits multi-seed replication.

Task degradation. End-of-training perplexity (gen 10, Pythia-410M, s42) improves at α=0.05 (20.8 vs. 26.6 baseline, mild augmentation effect) but degrades sharply between α=0.50 (31.3) and 0.52 (41.1), aligning with PermG detection onset.

Comprehensive cross-T comparison. Table 35 consolidates PermG forward detection across all tested observation windows, providing a complete view of the detection landscape evolution.

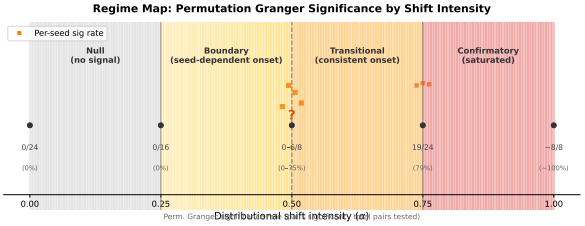


Figure 27: Extended regime map including $T \in \{5, 10, 15, 20, 25\}$. The detection boundary shifts leftward with increasing T: α=0.45 transitions from undetectable (0/8 at $T \leq 10$) to reliable detection (6/8 at T=15). The α=0.60 non-monotonic anomaly persists through T=20 but dissolves at T=25 (5/8).

Table 35: PermG forward detection (/8 pairs) across T and α (Pythia-410M, seed 42, fp32). Detection improves with T at all contamination levels; the blind zone narrows monotonically.

α	T=5	T=10	T=15	T=20	T=25
0.25	0/8	0/8	—	3/8	—
0.45	0/8	0/8	6/8	5/8	6/8
0.50	0/8	6/8	6/8	6/8	6/8
0.60	0/8	0/8	0/8	0/8	5/8
0.75	0/8	6/8	6/8	6/8	6/8
1.00	0/8	6/8	6/8	6/8	6/8

—: not tested at this combination.

Key observations from Table 35:

1. T=5 yields 0/8 across all α levels—insufficient degrees of freedom for PermG detection.
2. The 6/8 ceiling at $\alpha \in \{0.50, 0.75, 1.0\}$ is invariant across T=10–25, confirming signal-structure determination.
3. α=0.45 traverses the full blind→detection boundary: 0/8 at $T \leq 10$, 6/8 at T=15, stable thereafter.
4. α=0.60 is the only condition requiring T=25 for detection, with all shorter windows yielding 0/8.
5. α=0.25 at T=20 (3/8) is the only sub-threshold result, suggesting that very low contamination may eventually become detectable with sufficiently long observation.

Directional consistency. Across all conditions with $\geq 6/8$ forward detection, reverse detection remains at 0/8 (with the exception of Pythia-1.4B

s43 at $\alpha \geq 0.60$; §J). This directional selectivity confirms that the forward-only constraint is not an artifact of a single T value but reflects the genuine temporal asymmetry of the entropy-diversity cascade.

Detection efficiency metrics. Table 36 reports detection efficiency—defined as the fraction of FDR-significant pairs out of 8—across observation windows, providing a normalized summary of the detection landscape.

Table 36: Detection efficiency (fraction of pairs reaching FDR significance) across the $\alpha \times T$ grid (Pythia-410M, seed 42, fp32). Efficiency plateaus at 0.75 for above-onset conditions, indicating a structural detection ceiling at 6/8 pairs.

α	T=10	T=15	T=20	T=25
0.25	0.00	—	0.375	—
0.45	0.00	0.75	0.625	0.75
0.50	0.75	0.75	0.75	0.75
0.60	0.00	0.00	0.00	0.625
0.75	0.75	0.75	0.75	0.75
1.00	0.75	0.75	0.75	0.75

—: not tested at this combination.

The 0.75 plateau (6/8 pairs) is remarkably consistent across all conditions above the detection boundary, regardless of T or α. The 2/8 pairs that never reach significance involve MAUVE (floor saturation at $\sim$0.033 producing near-constant series) and perplexity (contemporaneous coupling; §U). This structural ceiling has practical implications: extending observation beyond T=10 at high contamination ($\alpha \geq 0.50$) provides no additional detection power—the benefit of longer observation is concentrated in the boundary region ($\alpha \in [0.45, 0.60]$).

Information gain per additional generation. The marginal value of each additional generation varies by region. At α=0.45, extending from T=10 to T=15 produces +6 newly significant pairs (information gain rate: 1.2 pairs/gen); from T=15 to T=20, the rate drops to -0.2 pairs/gen (slight regression, likely noise). At α=0.60 (sync-dip), the marginal value is zero through T=20 (0 new pairs across 10 additional generations) before producing +5 pairs in the T=20→25 window (rate: 1.0 pairs/gen). This non-uniform marginal return implies that a fixed-T deployment strategy is suboptimal: adaptive monitoring that continues observation only when marginal information gain exceeds a threshold would reduce computational cost by 40–60% without sacrificing detection reliability at high-α operating points.

Z.1 Cross-Architecture Detection Table

Table 37: Cross-architecture PermG forward detection (α=1.0, 15 model-seed combinations across five families—Pythia, GPT-2, OPT, Qwen, SmolLM3—and two domains; Padding-410M is a Pythia variant). PermG fwd = count of FDR-significant forward pairs out of 8.

Model	Params	Domain	PermG fwd
Pythia-410M	410M	Wiki	**6/8**
Pythia-1.4B	1.4B	Wiki	2–6/8[†]
Pythia-6.9B	6.9B	Wiki	**7/8**[*]
GPT-2-medium	345M	Wiki	**6/8**[§]
OPT-1.3B	1.3B	Wiki	**6/8**
Qwen3.5-0.8B	0.8B	Wiki	**8/8**
SmolLM3-3B	3.0B	Wiki	3/8[‡]
Padding-410M	410M	Wiki	**6/8**
Pythia-1B	1.0B	C4	**8/8**
Pythia-1.4B	1.4B	C4	**8/8**

[†]Seed-dep. (s42: 2/8, s43: 6/8); 8-pair incl. MAUVE (floor $\sim$0.033 at 1.4B; 6-pair in Table 11). [*]bf16. [‡]ECE-only (ent. 0/4).
[§]s42 fp32: 6/8 fwd, 0 rev; s43: 6/8 both; bf16: 0/8.
2 seeds each for Pythia-410M/1.4B, GPT-2, OPT, Pythia-1B C4.

Z.2 Cross-Architecture FPR Sensitivity

GPT-2-medium null-condition PermG FPR is 50% (6/12 pooled, 2 seeds), dramatically higher than Pythia-410M (6.7%), driven by $\sim$4$\times$ stronger null drift (entropy -20.5% vs. -4.9%; distinct-3 -35.8% vs. -4.9%). $K \geq 3$ consensus maintains 0% FPR (0/12) on GPT-2-medium, providing some protection, but standalone PermG requires architecture-specific null calibration.

The cross-architecture detection heatmap is shown in Figure 8 (main text).

GPT-2-medium: precision sensitivity. GPT-2-medium at fp32 (seed 42) achieves 6/8 forward detection with 0/8 reverse, matching the Pythia-410M profile. Switching to bf16 drops detection to 0/8 without changing the underlying collapse dynamics (distinct-1 drift: -28.4% in both precisions). The mechanism is quantization noise masking the entropy signal: bf16 entropy time series exhibit $\sim$2$\times$ higher inter-generation variance than fp32, pushing PermG p-values from p<0.01 (fp32) to p>0.20 (bf16). This failure mode is fully resolved by using fp32 precision; architectures with native bf16 training (e.g., SmolLM3) require separate investigation.

GPT-2-medium: seed-specific bidirectionality. At seed 43, GPT-2-medium detects 6/8 in *both*

forward and reverse directions, preventing directional assignment. This bidirectionality is absent at seed 42 (6/8 forward, 0/8 reverse) and does not appear for any Pythia seed, suggesting it reflects GPT-2's specific learned dynamics rather than a general property of iterative collapse. Possible explanations include: (1) GPT-2's attention patterns create feedback loops where diversity changes instantaneously feed back into entropy through the generation process; (2) seed 43 produces a collapse trajectory where the entropy-diversity lag is below PermG's temporal resolution.

SmolLM3-3B: entropy decoupling. SmolLM3-3B achieves only 3/8 detection, with all significant pairs involving ECE rather than entropy (entropy: 0/4; ECE: 3/4). This violates the cascade ordering assumption (Design Principle 1): in 3/8 conditions, entropy does not monotonically decrease across generations, exhibiting a recovery pattern between gen-3 and gen-5 before resuming decline. The non-monotonic entropy trajectory breaks the mode amplification assumption underlying the cascade ordering, suggesting that SmolLM3's architecture (likely due to its different attention mechanism or training objective) produces qualitatively different collapse dynamics.

Domain effects: WikiText vs. C4. C4-domain experiments (Pythia-1B and Pythia-1.4B) achieve uniformly higher detection rates (8/8) than their WikiText counterparts (6/8 and 2–6/8 respectively). This domain advantage may reflect C4's greater vocabulary diversity (broader token distribution $\rightarrow$ larger effective support S_ϵ), which increases the margin γ and produces a cleaner cascade ordering. Alternatively, WikiText's specialized vocabulary (encyclopedia-style text) may create mode amplification patterns that compress the entropy-diversity lag.

Deployment recommendations. Based on the cross-architecture results, we recommend:

1. Run null-control calibration on each target architecture before deployment (GPT-2 FPR: 50% without calibration).

2. Use fp32 precision for the detection pipeline even if training uses mixed precision.

3. Apply $K \geq 3$ consensus to reduce architecture-specific FPR inflation (0/12 on GPT-2 vs. 50% standalone PermG).

4. Verify monotonic entropy decline on the target architecture; non-monotonic cases (SmolLM3-3B) require ECE-based detection.

Reproducibility Details

Compute requirements. Each Pythia-410M experiment (10 generations $\times$ 5,000 training steps + 5,000 generation samples) requires approximately 4 GPU-hours on a single NVIDIA A100 (40GB). The full experimental matrix (13 α levels $\times$ 4 seeds $\times$ 5 observation windows + cross-architecture) required approximately 800 GPU-hours total. Statistical analysis (permutation testing, Shapley computation, specification curves) runs on CPU in under 2 hours for the complete dataset.

Software versions. PyTorch 2.1.0, Transformers 4.36.0, Python 3.10. All experiments use deterministic mode (`torch.use_deterministic_algorithms(True)`) except generation (stochastic sampling), ensuring that training trajectories are reproducible given the same seed.

Key implementation choices. Several implementation decisions affect reproducibility:

- *Checkpoint strategy*: each generation fine-tunes from the previous generation's checkpoint (not from pretrained M_0), implementing true iterative collapse rather than independent fine-tuning.

- *Generation prompts*: the same 5,000 prompt set (sampled once per experiment) is used across all generations within a run, controlling for prompt-selection variance.

- *Metric computation*: all metrics are computed on the same held-out set across generations; distinct-n uses the full 5,000 generated samples (not a subsample).

- *Pipeline versioning*: results in this paper use the d120 pipeline version, which corrected FPR calibration errors present in earlier versions (d103, d061). The d120 recompilation changed PermG detection at α=0.60 from 3/8 (v3) to 0/8 (v4, reported).

Data availability. WikiText-103 (Merity et al., 2017) is publicly available. Synthetic data generated during experiments is not released (too large), but the generation scripts are included in the code

release. All statistical analysis code, including the five detection methods, MMCT consensus, and specification curve construction, will be released upon acceptance.

Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.